
TECHNICAL REPORT

Calm-Trend Regime Strategies for BTC/USDT and ETH/USDT

Trend following with regime-based position sizing,
backtested on Binance hourly data, 2021–2025



Team Jain Streeter

Priyanshu Debnath
Kushagra Verma

Inter IIT Tech Meet 15.0

Inter IIT Bootcamp
Quant problem statement

Code, data pipeline and all result files: github.com/PriyanshuIITGHY2006/InterIIT_Quant_Jain-Streeter

Abstract

We designed two daily trading strategies from one rule set and backtested them on Binance hourly candles for 2021–2025 with 10,000 USDT, 0.15% cost per fill and next-open execution. Our data research found that the direction of BTC and ETH is close to unpredictable at useful horizons, while the size of their moves is strongly predictable. The strategies therefore use a slow, averaged trend signal to decide whether to hold and the market's volatility and trend-quality state to decide how much. CTR-S (BTC) adds a half-size short only in calm, fully confirmed bear markets; CTR-ETH (ETH) is long only. Over 2021–2025, CTR-S returned +368.4% (buy-and-hold +197.9%) with a maximum drawdown of −29.6% (−76.6%), and CTR-ETH returned +481.7% (+306.5%) with −37.3% (−79.3%). This report documents the data analysis, the indicator, machine-learning and strategy research, the mathematics behind each rule, every idea we tested and rejected, the backtest engine, the risk plan and the results.

Contents

1	Introduction	3
1.1	The task	3
1.2	Approach in one paragraph	3
1.3	Headline results, 2021–2025	3
2	Data and cleaning	3
2.1	Inventory	3
2.2	Quality checks	4
2.3	Cleaning pipeline	4
2.4	A structural break made by the exchange	4
3	Exploratory data analysis	5
3.1	Distribution of returns	5
3.2	Behaviour changes through time	6
3.3	Memory: direction has none, size has a lot	7
3.4	BTC and ETH	8
3.5	Seasonality	9
3.6	Informative versus redundant variables	9
3.7	Robust versus fragile relationships	9
3.8	What the data told us	11
4	Indicator research: why direction fails and volatility works	11
4.1	Study design	11
4.2	The model behind the verdicts	11
4.3	The information coefficient and the cost hurdle	12

4.4	Non-stationarity: an IC that changes sign	12
4.5	Why each family of directional indicators fails	13
4.6	Why the volatility side works	14
5	Machine-learning and reinforcement-learning research	17
6	Strategy development and rejected approaches	18
6.1	Process discipline	18
6.2	BTC: from baselines to CTR-S	19
6.3	Rejected approaches and why they failed	19
6.4	Break-even hit rate for short-term trading	20
6.5	ETH: why no ETH-specific design survived	21
7	The final strategy	21
7.1	The market-state framework	21
7.2	The daily decision	23
7.3	Why each layer is there	23
7.4	The BTC short side (CTR-S)	25
7.5	ETH (CTR-ETH)	26
7.6	Execution settings	26
8	Backtest engine and look-ahead control	27
8.1	Design	27
8.2	Keeping look-ahead out	27
9	Risk management	28
9.1	Payoff, not accuracy	28
9.2	Drawdown arithmetic and volatility drag	28
9.3	Tail risk and known risks	28
10	Results, 2021–2025	29
10.1	Year by year	31
10.2	Quarter by quarter	31
10.3	Robustness	32
10.4	How much of the result could be luck?	32
11	Limitations and further work	32
	References	33

1 Introduction

1.1 The task

The problem statement asks for a trading strategy on BTC/USDT and ETH/USDT, designed from data research, backtested over 1 January 2021 to 31 December 2025 and evaluated on 15 required metrics. The fixed conditions are a starting capital of 10,000 USDT, a cost of 0.15% of notional on every fill (fee and slippage, so a round trip costs 0.30%), and decisions that can only be executed after the information they use is available.

1.2 Approach in one paragraph

We studied the data before writing any strategy (Sections 2–3), tested more than 25 indicators and a set of machine-learning and reinforcement-learning models (Sections 4–5), and developed the strategy through a sequence of pre-registered stages in which most ideas failed (Section 6). The final design (Section 7) separates the two jobs a position has: a robust trend signal decides *whether* to be long, flat or short, and the market’s measured state decides *how much*. The same code is the BTC and the ETH strategy; every threshold is a rank within the coin’s own history, so nothing was tuned per coin.

1.3 Headline results, 2021–2025

	BTC		ETH	
	CTR-S	Buy-and-hold	CTR-ETH	Buy-and-hold
Total return	+368.4%	+197.9%	+481.7%	+306.5%
Sharpe ratio	0.98	0.67	0.96	0.75
Maximum drawdown	−29.6%	−76.6%	−37.3%	−79.3%
Worst calendar year	−4.3%	−64.2%	−20.2%	−67.5%
	(2025)	(2022)	(2022)	(2022)
Quarters beating buy-and-hold	11 of 20	–	10 of 20	–
Down quarters beating buy-and-hold	8 of 9	–	7 of 8	–

Table 1: Summary of the required 2021–2025 backtest. All result files are in [results/](#); we re-ran both strategies from the raw candles and reproduced every file exactly.

2 Data and cleaning

2.1 Inventory

The source is Binance spot klines from the public REST endpoint `/api/v3/klines`, for BTC/USDT and ETH/USDT, hourly, from 2021-01-01 00:00 to 2025-12-31 23:00 UTC. Each candle carries open, high, low, close, base volume, quote volume, number of trades and taker-buy volume. The raw files have 43,810 rows each; after completing the time grid there are 43,824 hours per coin. Daily, 4-hour and processed hourly files are derived from the cleaned hourly data and split into train (2021–2023), validation (2024) and test (2025).

A candle is labelled by its *open* time. Its open price is known at that moment; its high, low, close and volume only when it closes. A decision computed from bar t can be executed no earlier than

the open of bar $t + 1$.

2.2 Quality checks

Check	Result
Missing hours	14 hours in 7 outages (longest 4 hours on 2021-08-13), identical in both coins
Hidden outage bars	3 bars with zero or almost no trading (2021-02-11 03:00, 2021-04-25 04:00, 2023-03-24 12:00); the two with zero trades are flagged as outages
OHLC consistency, VWAP inside range, taker $\leq$ volume, positive prices	all pass, 0 violations
Open versus previous close	median gap $\approx 10^{-7}$; only 5 jumps above 0.1% (max 0.34%)
Daily bars rebuilt from hourly versus Binance daily	exact match
Extreme moves	all genuine: both coins move together on $2\text{--}25\times$ normal volume; BTC-only spikes on 2021-01-29 (+11.6%) and 2021-02-08 (+8.6%) match the Musk and Tesla news
Huge wicks	genuine liquidation cascades (2021-05-19, 2021-12-04, 2025-10-10); kept

Table 2: Data quality checks ([notebooks/01_data_research.ipynb](#)). Verdict: the data is trustworthy.

2.3 Cleaning pipeline

Step	Rule	Why
Sort, deduplicate	keep the first row per timestamp	exchange exports occasionally repeat rows
Validate OHLC	prices positive; high and low widened to bound open and close	a high below the close breaks range-based volatility
Complete the grid	every hour from first to last timestamp	indicators assume a regular clock
Fill gaps forward only	flat candle at the previous close, zero volume	interpolation would use the next price, which is future data
Flag outages	<code>is_filled</code> for filled hours and for <code>trades == 0</code>	the engine never trades on them
Daily bars	<code>resample("1D", label="left", closed="left")</code>	a day is complete only at 00:00 UTC the next day
Realised measures	per day, from that day's hourly returns	intraday information known at the day's close

Table 3: Cleaning steps ([src/data/preprocess.py](#); the same rules for new data in [src/system/ingest.py](#)).

2.4 A structural break made by the exchange

During the zero-fee promotion BTC trades per day rose from 1.6 million to 6.3 million ($3.8\times$), volume doubled and the average trade size halved (1,687 to 870 USDT), while ETH, which had no promotion, declined. When the promotion ended BTC activity fell below its earlier level, and the average trade size of both coins fell sharply again in late 2024–2025. Absolute volume, trade count and trade size are therefore not stationary for reasons unrelated to price behaviour, and **no activity measure is used as a trading signal**.

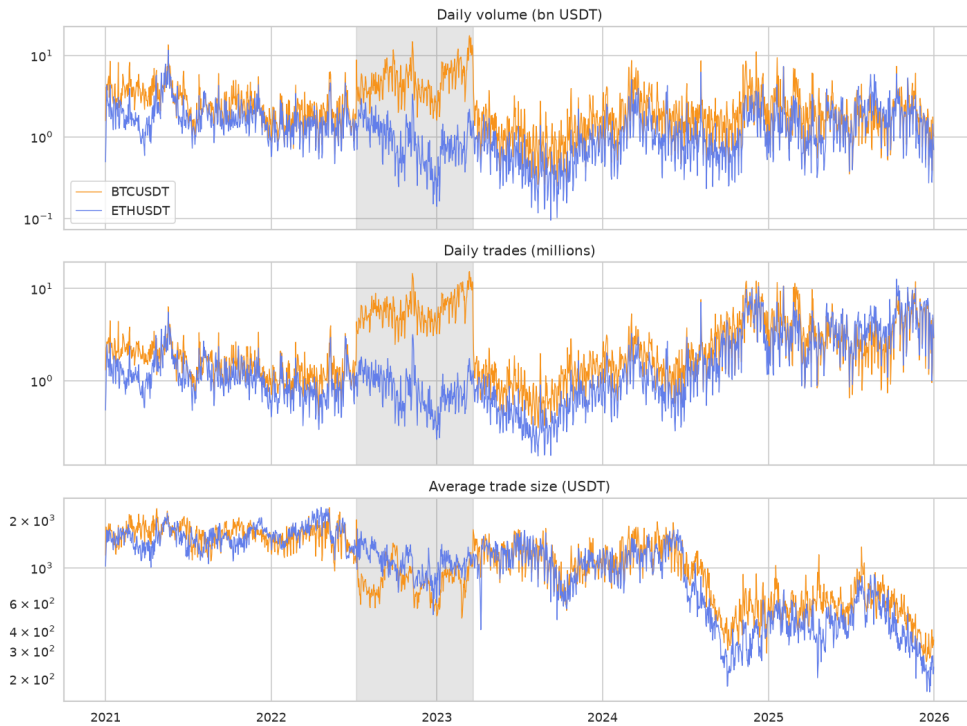


Figure 1: Daily volume, trades and average trade size. The shaded band is Binance's BTC zero-fee promotion (8 July 2022 – 22 March 2023).

3 Exploratory data analysis

All statistics in this section use only the training years 2021–2023, so 2024 and 2025 remained out-of-sample checks.

3.1 Distribution of returns

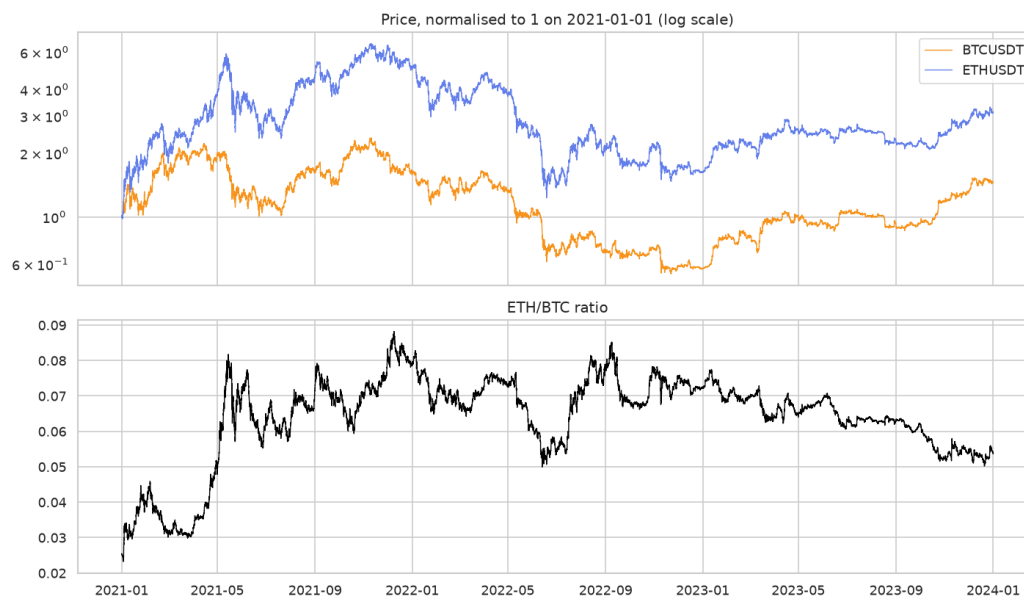
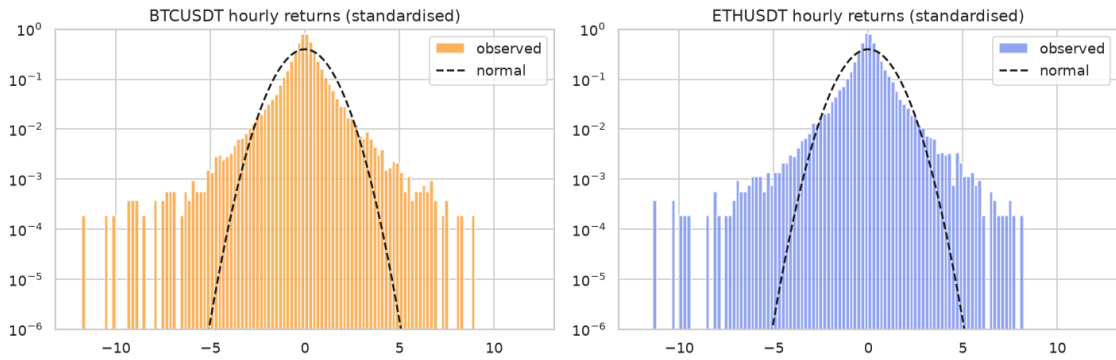


Figure 2: BTC and ETH normalised to 1 on 2021-01-01 (log scale), and the ETH/BTC ratio.

2021–2023	BTC	ETH
Annualised volatility	~65%	~85%
Hourly excess kurtosis	15.7	13.4
Daily excess kurtosis	3.5	5.7
Hourly skew	−0.19	−0.59
Tail index α (Hill, hourly)	2.7–3.7	2.6–4.0
Student- t degrees of freedom (daily)	2.5	2.9
Moves above 4σ , relative to a normal	~135×	~134×

Table 4: Distribution statistics.**Figure 3:** Standardised hourly returns against the normal density (log scale).

The Hill estimator of the tail index from the k largest absolute returns $X_{(1)} \geq \dots \geq X_{(k)}$ is

$$\hat{\alpha} = \left(\frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k)} \right)^{-1}. \quad (1)$$

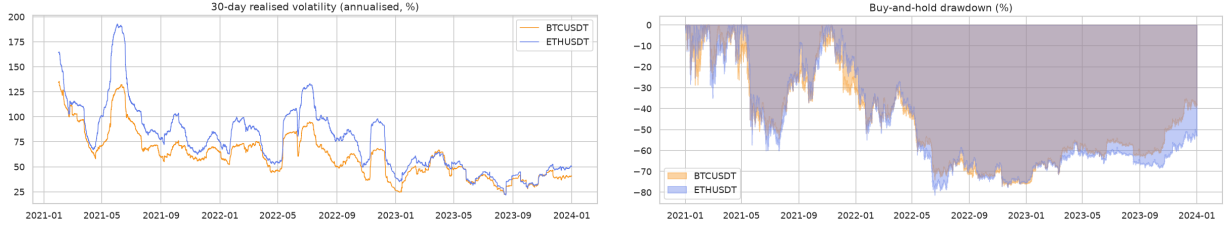
With $\hat{\alpha} \approx 3$ the variance is finite but $\mathbb{E}|r|^p < \infty$ only for $p < \alpha$, so the fourth moment is probably not. Kurtosis estimates jump as data is added, and every threshold in our strategies is therefore a rank, a percentile or a median, never a z-score. Log prices are non-stationary and returns are stationary in the mean (ADF and KPSS agree); return variance is not constant.

Discovery: jumps arrive in calm markets. Scaling hourly returns by past volatility normally lowers kurtosis by removing volatility clustering. Here it *raises* it (BTC 15.7 to 24.7), for EWMA spans of a day, a week and a month. The largest scaled shocks are 5% hourly moves that arrive when recent hourly volatility was about 0.25% (for example 2023-08-17 and 2023-08-29); removing five such hours halves the scaled kurtosis. Volatility-targeted sizing therefore puts on its largest positions exactly where these jumps hit, which is why our strategies cap exposure at $1.5\times$ regardless of how calm the market is, and why the engine fills stops at the open when a bar gaps through them.

3.2 Behaviour changes through time

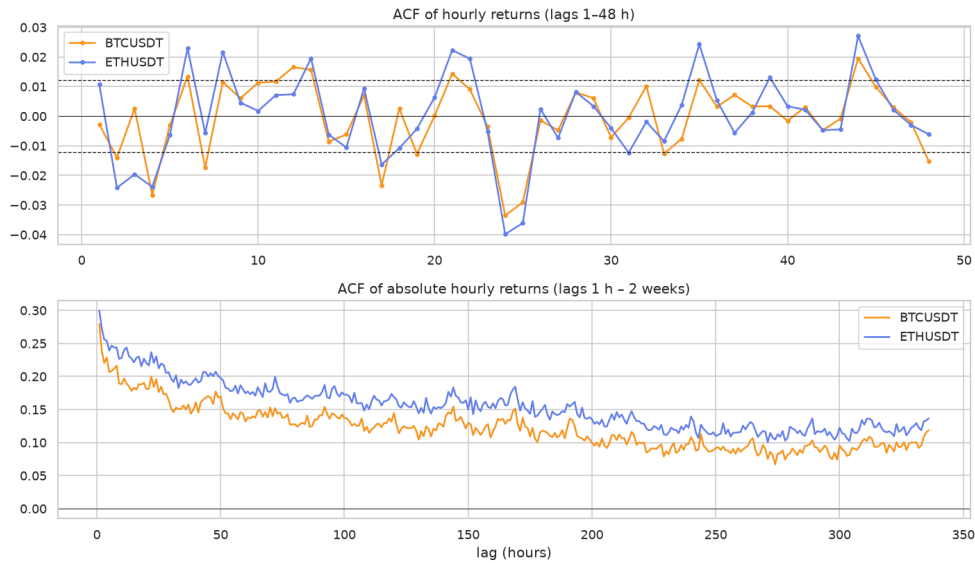
Volatility halved from 2021 to 2023 (30-day volatility ranged 22–135% for BTC and 22–193% for ETH) and skew turned from negative to positive. Buy-and-hold fell 77% (BTC) and 81% (ETH) and was still about 780 days under water at the end of 2023. The market’s character also changed: BTC’s weekly variance ratio was 0.83 (mean-reverting) in 2021, 1.05 in 2022 and 1.34

	BTC			ETH		
Year	Return	Volatility	Max DD	Return	Volatility	Max DD
2021	+58%	81%	−55%	+404%	109%	−60%
2022	−64%	64%	−67%	−67%	87%	−77%
2023	+156%	44%	−22%	+91%	47%	−28%

Table 5: Three training years, three different markets.**Figure 4:** Left: 30-day realised volatility. Right: buy-and-hold drawdown, 2021–2023.

(trending) in 2023. A single fixed style, pure trend or pure mean reversion, is exposed to these changes.

3.3 Memory: direction has none, size has a lot

**Figure 5:** Autocorrelation of hourly returns (lags 1–48 h) and of absolute hourly returns (lags up to two weeks).

The autocorrelation of returns is close to zero at all lags, and variance ratios are close to 1 at every horizon from 2 hours to 20 days. The variance ratio is a weighted sum of autocorrelations,

$$VR(q) = \frac{\text{Var}\left(\sum_{i=1}^q r_{t+i}\right)}{q \text{Var}(r_t)} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_k, \quad (2)$$

and the heteroskedasticity-robust Lo–MacKinlay estimates on daily BTC are 0.97, 1.00, 1.01 and 1.02 at $q = 2, 5, 10, 20$, all with $|z| < 1$. The size of moves, in contrast, has long memory: absolute returns stay autocorrelated for weeks with a 24-hour cycle, and the autocorrelation of log realised variance is 0.63–0.71 at one day and 0.26–0.33 at 90 days (Figure 6). Three further facts:

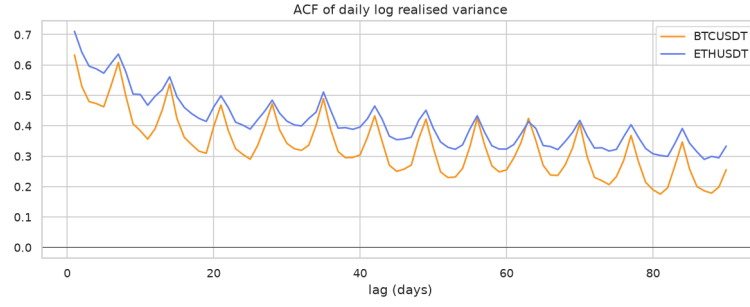


Figure 6: Autocorrelation of daily log realised variance.

- The Spearman lag-1 autocorrelation is -0.04 to -0.05 while Pearson is ≈ 0 : ordinary moves slightly reverse, the few huge ones do not, and they dominate Pearson.
- There is no leverage effect. In GJR-GARCH(1,1), $h_t^2 = \omega + \alpha r_{t-1}^2 + \gamma r_{t-1}^2 \mathbf{1}\{r_{t-1} < 0\} + \beta h_{t-1}^2$, the asymmetry term is insignificant ($p = 0.95$ BTC, 0.68 ETH): crypto volatility reacts about equally to up and down moves.
- Fitted GARCH persistence is close to 1. This reflects the shifting volatility level rather than permanent shocks; one GARCH is mis-specified over this sample.

Volatility is forecastable out of sample. Corsi’s heterogeneous autoregressive (HAR) model approximates long memory with three horizons,

$$\ln RV_{t+1} = \beta_0 + \beta_d \ln RV_t^{(1)} + \beta_w \ln RV_t^{(5-7)} + \beta_m \ln RV_t^{(22-30)} + u_{t+1}. \quad (3)$$

Fitted on 2021–22 (BTC: $\hat{\beta}_d = 0.29$, $\hat{\beta}_w = 0.51$, $\hat{\beta}_m = 0.13$; ETH: 0.37 , 0.47 , 0.06), it explains 60% (BTC) and 73% (ETH) of next-day log volatility in 2023, with 34% lower mean squared error than the forecast “tomorrow equals today”. The weekly term dominates.

3.4 BTC and ETH

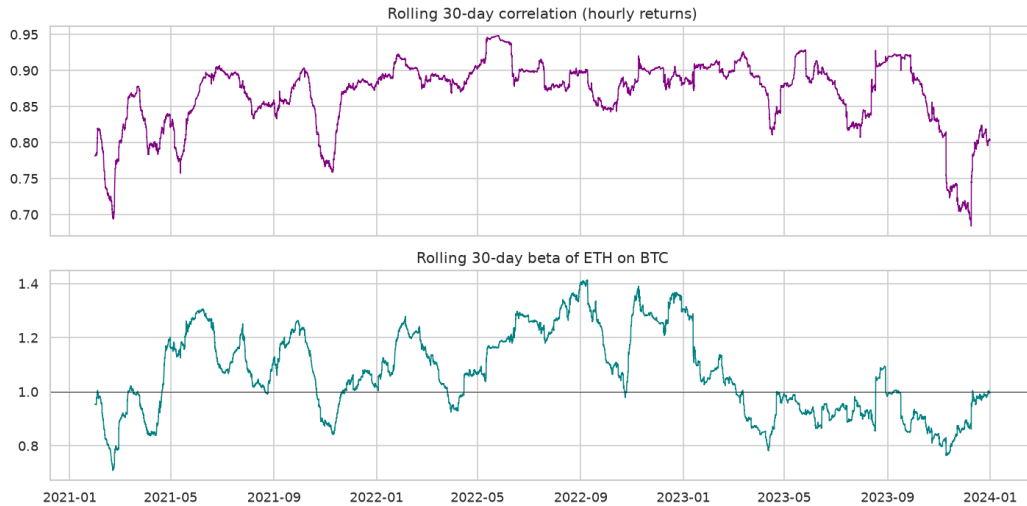


Figure 7: Rolling 30-day correlation of hourly returns and rolling beta of ETH on BTC.

Dependence is asymmetric in the tails. For a quantile q define the lower co-exceedance probability

$$\lambda_L(q) = \mathbb{P}(r^{ETH} \leq F_{ETH}^{-1}(q) \mid r^{BTC} \leq F_{BTC}^{-1}(q)) \quad (4)$$

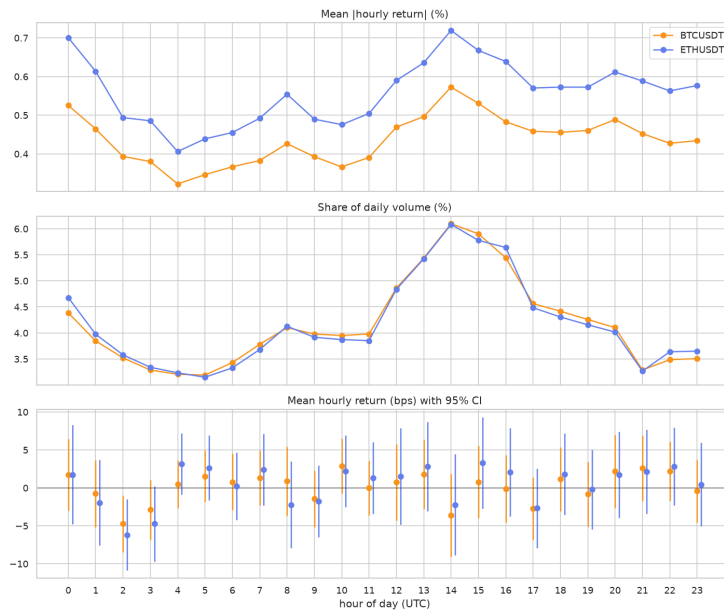
Correlation (1 h / 4 h / 1 d / 1 w)	0.85 / 0.85 / 0.82 / 0.83
Rolling 30-day correlation range	0.68 – 0.95
ETH beta to BTC (mean, range)	1.07 (0.71 – 1.41)
Share of ETH variance explained by BTC	68–73%
Lead–lag at ± 1 hour	≈ 0

Table 6: BTC–ETH co-movement.

and $\lambda_U(q)$ with the inequalities reversed. On the worst and best 1% of days $\lambda_L = 0.82$ and $\lambda_U = 0.27$, against 0.41 and 0.42 for a bivariate normal with the same correlation; the hourly figures (263 observations) point the same way (0.65 and 0.50). The coins crash together and rally apart, so holding both gives diversification in rallies and none in crashes.

ETH/BTC “mean reversion” looked significant on the full sample (ADF $p = 0.017$, cointegration $p = 0.054$) but had ADF $p \geq 0.19$ in every single year: the full-sample result is one round trip of the ratio. We dropped pairs trading.

3.5 Seasonality

**Figure 8:** Mean absolute hourly return, share of daily volume, and mean hourly return with 95% intervals, by hour of day (UTC).

Volatility and volume follow a stable daily cycle (peak at 14–15 UTC, about 30% above average; trough at 04–06 UTC, about 25% below), with year-to-year profile correlation 0.65–0.71. Returns do not: the hourly return profile correlates -0.16 between years, one of 24 hours has $|t| > 2$ (1.2 expected by chance), and no weekday has $|t| > 1.1$. We use no calendar rule.

3.6 Informative versus redundant variables

3.7 Robust versus fragile relationships

Robust. Volatility clustering and its forecastability; the intraday volatility and volume cycle; fat tails and asymmetric crash dependence; and, statistically, short-term reversal: 33 of 78 predictive tests have $|t| > 2$ (3.5 expected by chance) and 17 have $|t| > 3$ (0.2 expected), with the same sign

Variable	Informative about		Strength and notes
Last-hour range	high/low	future volatility	IC 0.36–0.39 with the next 24 h absolute move, stable every year; best single variable
Past 24 h realised volatility		future volatility	IC 0.29–0.35, stable
Daily high/low (Parkinson)		next-day volatility	correlation 0.58–0.65, against 0.38–0.42 for the squared close return
Own 1–24 h returns		next 1–4 h direction (reversal)	IC -0.04 to -0.07 , t up to -11 ; economically too small (Section 3.7)
Taker buy ratio		same-hour return (mechanical)	next hour -0.03 to -0.04 ; mostly a proxy for the own return
Volume surprise		future volatility	IC 0.10–0.12 overall, but the sign flips between years
log_volume	vs	—	rank correlation 0.92: redundant
log_trades			
Amihud illiquidity		—	rank correlation 0.68 with $ r $: mostly the absolute return in disguise

Table 7: What each variable carries information about.

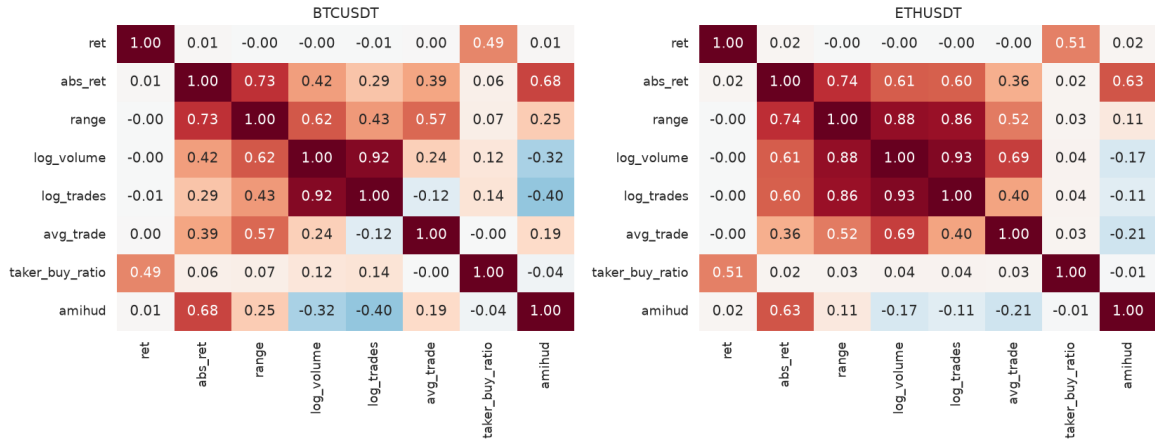


Figure 9: Rank correlations between the hourly variables.

in all three years.

Real but economically insufficient. The reversal is far smaller than the 30 bps a round trip costs:

Signal $\rightarrow$ target	Extreme-decile spread	Round-trip cost
BTC 4 h return $\rightarrow$ next 1 h	2.3 bps	30 bps
BTC 4 h return $\rightarrow$ next 4 h	12.9 bps	30 bps
ETH 4 h return $\rightarrow$ next 4 h	14.6 bps	30 bps

Table 8: The short-term reversal against costs.

Fragile. ETH/BTC mean reversion (above); daily time-series momentum, where a 30-day lookback shows BTC +64% a year but its 20-, 45- and 90-day neighbours are much weaker or negative and no lookback reaches $|t| = 2$; volume surprise as a volatility predictor (sign flips by year); hour-of-day return effects; and BTC mean reversion in high-volatility regimes (a single test at $t \approx -2.6$).

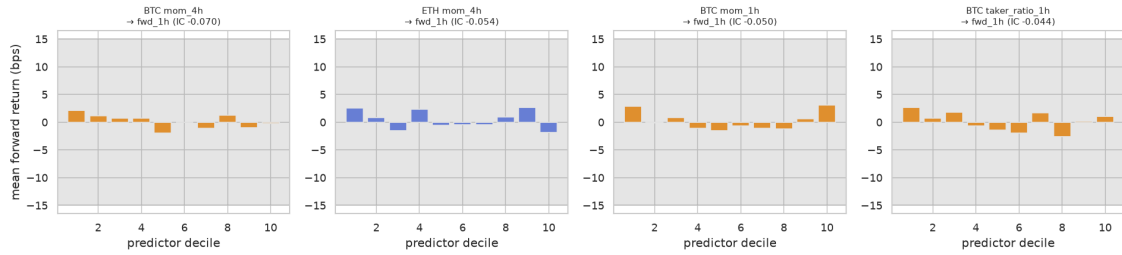


Figure 10: Mean forward return by decile of the predictor.

3.8 What the data told us

1. **It is a volatility market more than a direction market.** The size of the next move is far more forecastable than its sign; sizing and risk management are where the clearest edge lies.
2. **Two coins, one factor.** $\text{ETH} \approx 1.07 \times \text{BTC}$ plus independent noise, and the pair behaves as one asset in a crash.
3. **Regimes dominate.** Three years gave a bull, a bear and a recovery market with different volatility and different trend or reversion character.
4. **Danger comes out of calm.** The largest shocks relative to recent volatility arrive in quiet markets.
5. **Costs are large relative to short-horizon edges.** 30 bps per round trip removes every hourly effect found, so a viable strategy trades rarely and holds for days to months.
6. **Below the 200-day average,** next-day returns do not differ ($|t| \leq 1.1$) but volatility is higher (BTC 3.3% against 2.7% daily): a trend filter is a *risk* switch, not a return signal.

4 Indicator research: why direction fails and volatility works

4.1 Study design

[notebooks/02_indicator_analysis.ipynb](#) tested more than 25 indicators in six families (volatility estimators, trend strength and choppiness, oscillators, squeezes, flow and liquidity, and filters and regime models from the literature). Each was asked one question: *does what it says hold in every year, for both coins?* Predictive strength was measured by the rank information coefficient with next-week return, next-week volatility and next-week trend profit, year by year, with Newey–West standard errors for the overlapping 7-day targets.

4.2 The model behind the verdicts

Let P_t be the daily close, $r_t = \ln P_t - \ln P_{t-1}$, and $\mathcal{F}_t$ the information available at the close of day t . Write

$$r_{t+1} = \mu_t + \sigma_t \varepsilon_{t+1}, \quad \mathbb{E}[\varepsilon_{t+1} | \mathcal{F}_t] = 0, \quad \text{Var}(\varepsilon_{t+1} | \mathcal{F}_t) = 1. \quad (5)$$

Section 3 showed that σ_t is highly predictable and that $\mathbb{E}[r_{t+1} | \mathcal{F}_t] \approx \mu$, a constant. Every successful component of our strategy uses σ_t ; every failed indicator tried to predict μ_t .

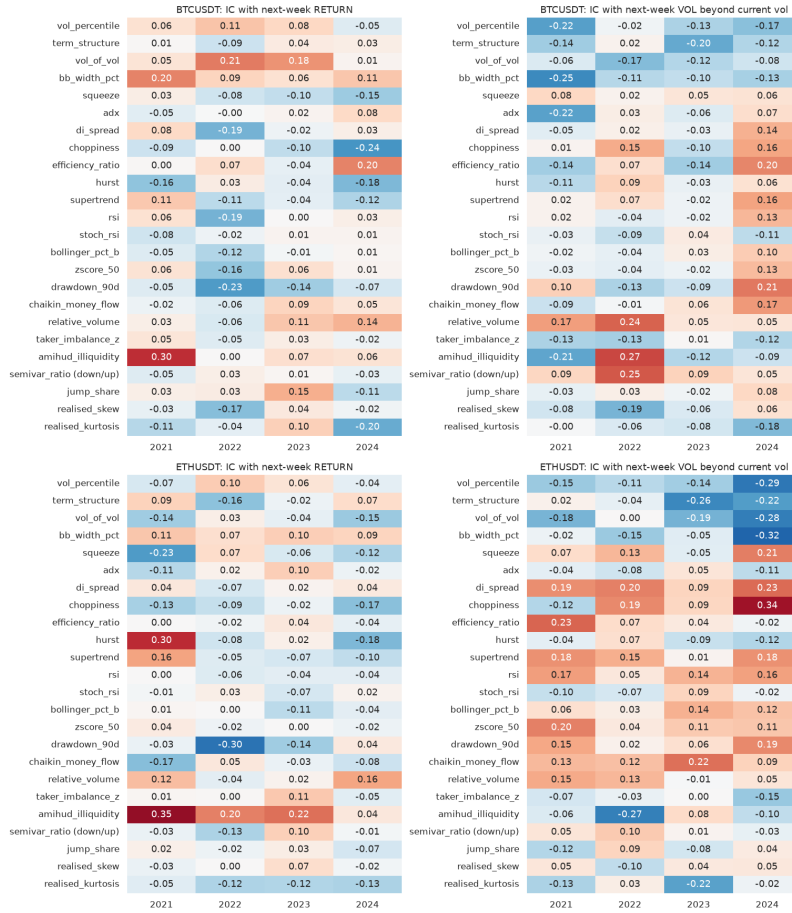


Figure 11: Indicator scorecard: yearly information coefficients of each indicator with next-week return (left) and with next-week volatility beyond current volatility (right), BTC (top) and ETH (bottom).

4.3 The information coefficient and the cost hurdle

Let s_t be an $\mathcal{F}_t$ -measurable signal with information coefficient $\rho = \text{corr}(s_t, r_{t+1:t+h})$. If (s, r) is close to jointly Gaussian with $\mathbb{E}[r] = 0$, the sign bet earns

$$\mathbb{E}[\text{sign}(s) r] = \mathbb{E}[\text{sign}(s) \mathbb{E}[r | s]] = \rho \frac{\sigma_r}{\sigma_s} \mathbb{E}|s| = \rho \sigma_r \sqrt{2/\pi}. \quad (6)$$

Our strongest short-horizon effect, the 4-hour return predicting the next 4 hours, has $\rho \approx -0.07$. With $\sigma_{4h} = 0.665/\sqrt{2190} \approx 1.42\%$ the gross edge is about 7.9 bps per bet, against 30 bps of round-trip cost; the measured extreme-decile spread (13–15 bps) confirms the order of magnitude. At portfolio level the fundamental law of active management, $\text{IR} \approx \text{IC}\sqrt{\text{BR}}$, gives the same verdict: a weekly signal has breadth of about 52 bets a year, so $|\text{IC}| \leq 0.05$ gives $\text{IR} \leq 0.36$ before costs. More generally, N round trips a year with gross edge e per trade earn roughly $N(e - 0.30\%)$.

4.4 Non-stationarity: an IC that changes sign

Treat each year's IC as $\text{IC}_y = \bar{\kappa} + u_y$ with $u_y \sim (0, \tau^2)$. For RSI, stochastic RSI, %B, z-score, ADX, DI spread, Supertrend, EMA ribbon, Aroon, Heikin-Ashi, Kalman slope and CUSUM the yearly ICs change sign (Supertrend: +0.11, -0.11, -0.04, -0.12), so $\tau \gg |\bar{\kappa}|$ and a signal fitted in one regime is a coin flip in the next.

Family	Verdict	Evidence
Volatility level and forecasts	reliable	persistent every year; range-based estimators agree within a few percent
Downside semivariance	reliable	about $2\times$ the forecasting weight of upside semivariance for BTC
Bipower variation	reliable	jumps fade within days; smooth volatility persists
Bollinger–Keltner squeeze	reliable for volatility only	volatility rises 10–14% after a squeeze; the break direction is random
Choppiness < 38.2, efficiency ratio	reliable as a trend-quality filter	trend payoff positive in clean trends in 4 of 4 years
RSI, stochastic RSI, Bollinger %B, z-scores	unreliable for direction	the sign of the yearly IC flips; one exception: BTC bounced after $RSI < 35$ in 4 of 4 years
ADX	unreliable for direction	measures trend strength, not sign
Supertrend, EMA ribbons, Heikin-Ashi, Kalman	no edge beyond a moving average	all are delayed linear filters of price
Hurst exponent	rejected	BTC’s persistence matched shuffled data
CUSUM, Markov switching	variance classifiers	separate calm from stormy, not up from down
Amihud illiquidity	promising but caveated	$IC \geq 0$ in 8 of 8 coin-years, but driven by volume drift and the zero-fee period

Table 9: Indicator verdicts. The design rule that came out of this study: use reliable indicators to decide size, and the simplest robust trend measure to decide direction.

4.5 Why each family of directional indicators fails

Oscillators. RSI is a bounded monotone transform of a ratio of Wilder averages, $RSI_t = 100 - 100/(1 + RS_t)$ with $RS_t = EWMA_{1/n}(r^+)_t / EWMA_{1/n}(r^-)_t$. It is $\mathcal{F}_t$ -measurable, so under the martingale-difference property $\mathbb{E}[r_{t+1} | RSI_t] = \mathbb{E}[\mathbb{E}[r_{t+1} | \mathcal{F}_t] | RSI_t] \approx \mu$ whatever its level. Overbought/oversold trading presumes $\rho_k < 0$ at the oscillator’s horizon; with $VR \approx 1$ there is none, and in trending years the sign reverses: $RSI > 70$ was followed by gains in 2023–24 and losses in 2022.

ADX. $DX_t = 100 |DI_t^+ - DI_t^-| / (DI_t^+ + DI_t^-)$ is unchanged when the price path is reflected (up-moves and down-moves swap, so $DI^+ \leftrightarrow DI^-$). It is an even function of direction and cannot carry directional information. Wilder smoothing is an EWMA with $\alpha = 1/14$ and mean lag 13 bars, applied twice, so ADX reports trend strength about 26 bars after the move.

Moving averages and their relatives. An n -day SMA has group delay $\tau_g = (n - 1)/2$ days (99.5 days for $n = 200$). To first order a linear trend rule holds $w_t = \sum_{k \geq 0} a_k r_{t-k}$, so its expected profit is a weighted sum of autocovariances,

$$\mathbb{E}[w_t r_{t+1}] = \sum_{k \geq 0} a_k \gamma(k + 1). \quad (7)$$

With $\gamma(k) \approx 0$ this is close to zero while every sign change of w costs $2c$. Fast filters flip more: a single 30-day rule made 59 trades in 2021–24 against 9 for the 100–250-day vote share, and its 50 trades shorter than 30 days lost 7,900 USDT. Slow filters pay only through rare, persistent autocorrelation episodes such as the multi-month 2023–24 trends.

Heikin-Ashi. With $HA_t^c = (O_t + H_t + L_t + C_t)/4$ and $HA_t^o = \frac{1}{2}(HA_{t-1}^o + HA_{t-1}^c)$, unrolling gives $HA_t^o = \sum_{j \geq 1} 2^{-j} HA_{t-j}^c$, an EWMA with $\alpha = 1/2$ of the lagged typical price. The candle colour is the sign of a very short momentum filter and inherits its zero expected profit.

Kalman filter. For the local-level model $x_t = \ell_t + \varepsilon_t$ ($\text{Var} = R$), $\ell_t = \ell_{t-1} + \eta_t$ ($\text{Var} = Q$), the prior variance converges to $P = (Q + \sqrt{Q^2 + 4QR})/2$ and the gain to $K = P/(P + R)$; the filtered level $\hat{\ell}_t = \hat{\ell}_{t-1} + K(x_t - \hat{\ell}_{t-1})$ is an EWMA. For our $Q/R = 0.01$, $K = 0.0951$: a 20-day exponential moving average.

CUSUM. Page's $S_t^+ = \max(0, S_{t-1}^+ + d_t - k)$ is optimal for detecting a shift between known densities of i.i.d. observations. Here $d_t = \ln P_t - \hat{\ell}_t$ is an EWMA residual with autocorrelation about $1 - K \approx 0.9$, and the mean shift is negligible relative to σ ; its bull/bear labels flip in predictive value year to year (ICs $+0.14, -0.11, +0.03, -0.08$).

Hurst exponent. For independent data the expected rescaled range is not $cn^{1/2}$ but (Anis-Lloyd)

$$\mathbb{E}[R/S]_n = \frac{n - \frac{1}{2}}{n} \frac{\Gamma(\frac{n-1}{2})}{\sqrt{\pi} \Gamma(\frac{n}{2})} \sum_{i=1}^{n-1} \sqrt{\frac{n-i}{i}}, \quad (8)$$

which over window sizes 8–64 implies a slope of 0.617 for pure noise. Our $\hat{H} = 0.589$ on BTC is indistinguishable from the 0.594 measured on shuffled BTC returns: the apparent persistence is estimator bias.

Markov switching and HMMs. The filtered recursion is $\alpha_t(j) \propto f_j(y_t) \sum_i \alpha_{t-1}(i) \Pi_{ij}$ and the predictive mean is $\sum_j (\sum_i \alpha_t(i) \Pi_{ij}) \mu_j$. Our fitted states have $\mu_j \approx 0.000$ and 0.001 per day but volatility levels of about 26% and 66% annualised, so the predictive mean is about zero whatever the probabilities. The model is a good volatility classifier (next-week volatility 0.59 against 0.47 across its states, persistence $\Pi_{11} = 0.83$, $\Pi_{22} = 0.90$) with no first-moment information.

PCA and K-means. Lloyd's algorithm minimises $\sum_i \|x_i - c_{k(i)}\|^2$. With power-law features the second moment is dominated by a few extreme observations, centroids absorb the tails, and 90.5% (BTC) and 93.5% (ETH) of days collapse into one cluster: the clustering separates outliers from everything else.

4.6 Why the volatility side works

Realised measures. For a jump-diffusion $d \ln P_t = \mu_t dt + \sigma_t dW_t + dJ_t$ and hourly returns $r_{t,i}$ of day t ,

$$RV_t = \sum_{i=1}^M r_{t,i}^2 \longrightarrow \int_{t-1}^t \sigma_s^2 ds + \sum_{t-1 < s \leq t} (\Delta J_s)^2, \quad (9)$$

$$BV_t = \frac{\pi}{2} \sum_{i=2}^M |r_{t,i}| |r_{t,i-1}| \longrightarrow \int_{t-1}^t \sigma_s^2 ds, \quad RS_t^\pm = \sum_i r_{t,i}^2 \mathbf{1}\{r_{t,i} \gtrless 0\}. \quad (10)$$

Bipower variation removes the jumps (Barndorff-Nielsen and Shephard), so $RV - BV$ isolates them, and the realised semivariances split RV by sign. Regressing next week's variance on the parts gives:

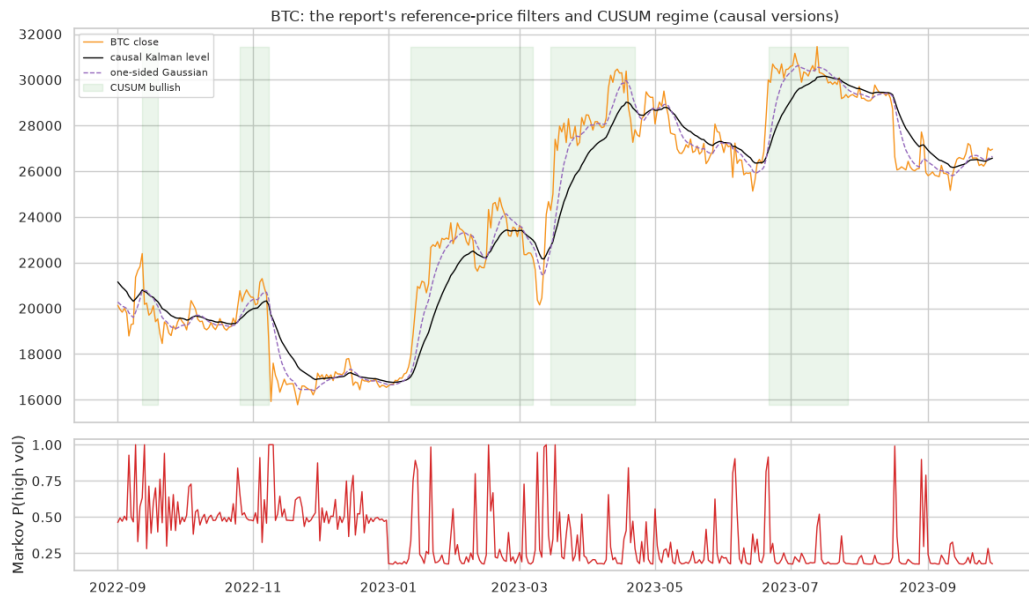


Figure 12: BTC with the reference report's price filters and CUSUM regime, in their causal versions (notebook 02).

Component	Coefficient on next week's variance
smooth (bipower)	0.58
jump	0.08
downside semivariance	0.45
upside semivariance	0.21

Table 10: Smooth risk persists, jumps fade, and bad volatility persists about twice as much as good.

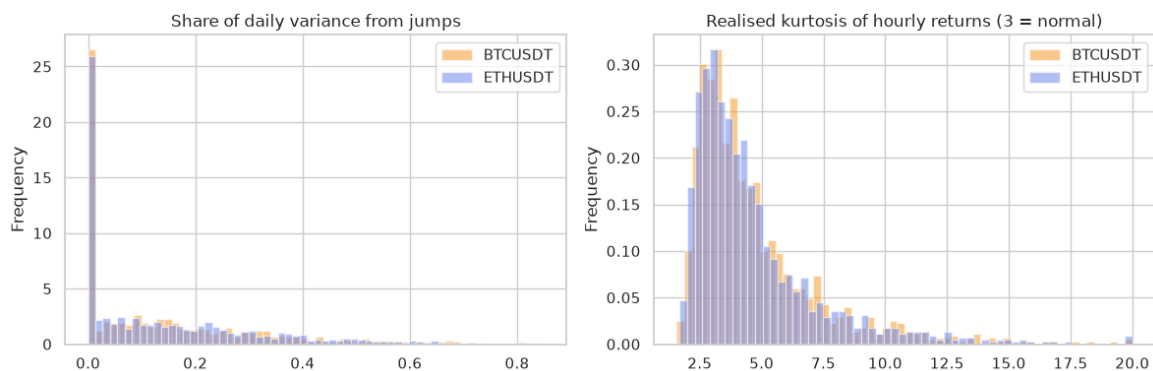


Figure 13: Share of daily variance from jumps, and realised kurtosis of hourly returns (notebook 02).

Range-based estimators. Parkinson's $\hat{\sigma}^2 = (\ln H/L)^2/(4 \ln 2)$ correlates 0.58 with next-day realised variance, against 0.38 for the squared daily return and 0.62 for hourly RV. Garman–Klass, Rogers–Satchell and Yang–Zhang estimates agreed within a few percent (Figure 14). This is why the strategy's volatility percentile uses Parkinson volatility and why it still works on daily candles alone.

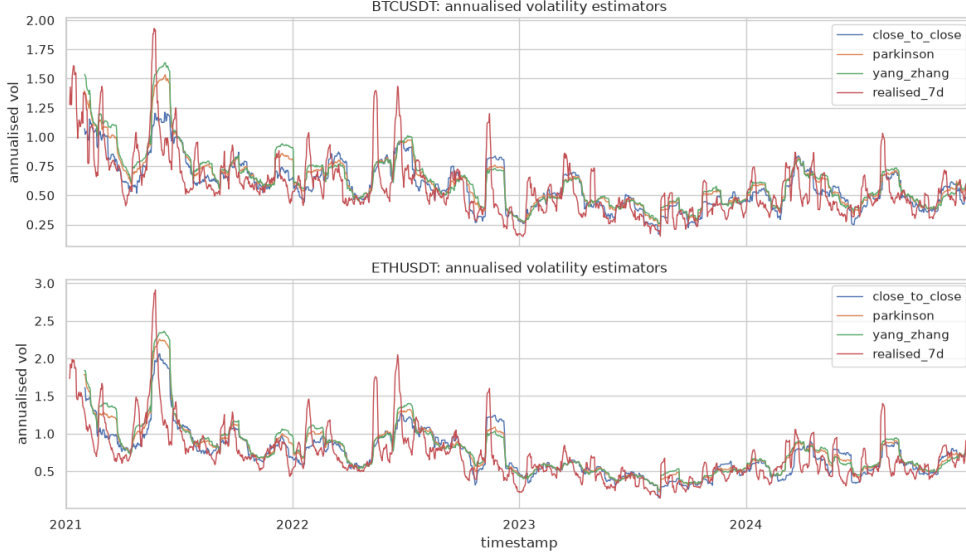


Figure 14: Annualised volatility estimators (notebook 02).

Proposition 1 (Volatility-managed exposure raises the Sharpe ratio). Assume $r_{t+1} = \mu + \sigma_t \varepsilon_{t+1}$ with σ_t known at t and ε i.i.d. $(0, 1)$ independent of σ_t . Constant exposure has $SR_0 = \mu/\sqrt{\mathbb{E}[\sigma_t^2]}$. For $w_t = c/\sigma_t$,

$$\mathbb{E}[w_t r_{t+1}] = c\mu \mathbb{E}[\sigma_t^{-1}], \quad \text{Var}(w_t r_{t+1}) = c^2(1 + \mu^2 \text{Var}(\sigma_t^{-1})).$$

On daily data $\mu^2 \text{Var}(\sigma^{-1}) \approx 0$, and Jensen's inequality applied to $x \mapsto 1/x$ and to $\sqrt{\cdot}$ gives

$$SR_{\text{managed}} \approx \mu \mathbb{E}[\sigma_t^{-1}] \geq \frac{\mu}{\mathbb{E}[\sigma_t]} \geq \frac{\mu}{\sqrt{\mathbb{E}[\sigma_t^2]}} = SR_0. \quad (11)$$

The condition is that μ is roughly independent of σ_t ; for BTC, mean returns do not differ across volatility terciles (all $|t| < 1.5$). In our ablation the volatility layer moved the Sharpe ratio from 0.99 to 1.06 and cut the maximum drawdown from -47% to -38% .

Trend quality against a random walk. The efficiency ratio $ER_n = |P_t - P_{t-n}| / \sum_{i=1}^n |\Delta P_{t-i+1}|$ has, for a driftless Gaussian walk, $\mathbb{E}|S_n| = \sigma\sqrt{2n/\pi}$ and $\mathbb{E}\sum |\Delta P| = n\sigma\sqrt{2/\pi}$, so $\mathbb{E}[ER_n] \approx 1/\sqrt{n} = 0.183$ at $n = 30$; simulation gives 0.183–0.187 and BTC averages 0.213. The choppiness index $CI_n = 100 \log_{10}(\sum TR/(\max H - \min L))/\log_{10} n$ tends to about 50 for a random walk; simulation gives 47.2 and BTC averages 49.7. Choppiness is BTC's normal state, and $CI < 38.2$ marks paths markedly straighter than chance (14% of days).

Trend pays only in calm, orderly markets. Splitting days by volatility and trend efficiency, a 30-day momentum bet earned money in calm trends in 4 of 4 years and roughly nothing in high-volatility regimes; on ETH the low-volatility trending regime paid $+2.3\%$ per week on average. Choppiness below 38.2 also paid in 4 of 4 years.

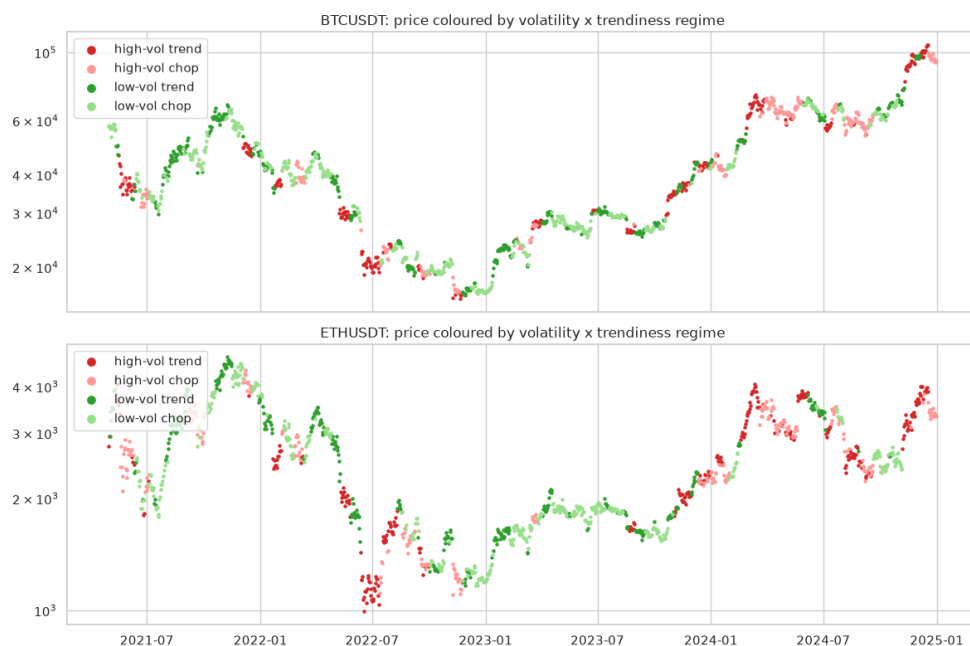


Figure 15: BTC and ETH coloured by four regimes: high-volatility trend, high-volatility chop, low-volatility trend, low-volatility chop (notebook 02).

The squeeze. When the Bollinger Bands sit inside the Keltner Channel, volatility rises 10–14% over the following days but the direction of the break is random, so the right response is to cut size, not to bet on a breakout.

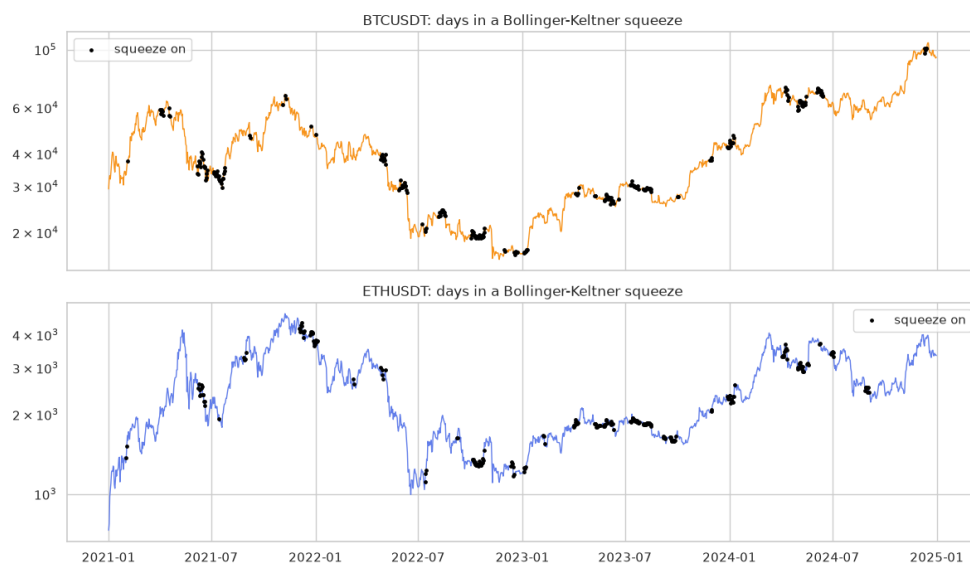


Figure 16: Days in a Bollinger–Keltner squeeze (notebook 02).

5 Machine-learning and reinforcement-learning research

A separate study tested whether machine learning could find what the rules missed. It used 2021–2023 data only, monthly walk-forward training with an 8–11 day embargo between training and test, settings fixed before any run, and shuffled-label placebos and constant-size controls.

Idea	Out-of-sample result (2022–2023)	Verdict
Direction classifiers (logistic regression, random forest, gradient boosting; 28 features)	AUC 0.40–0.51; placebo 0.52; trading the predictions lost 22% (BTC) and 82% (ETH)	dropped
HAR + gradient boosting for volatility	significantly worse than plain HAR for BTC (Diebold–Mariano $p = 0.005$), equal for ETH ($p = 0.87$)	HAR kept
Hidden Markov model regimes	cleaner calm/stormy separation, strategy unchanged (Sharpe 0.86 vs 0.88)	descriptive only
Triple-barrier trade filter	AUC 0.37–0.47; its gain equalled a constant half-size control	rejected
Tabular Q-learning	–79% (BTC), –83% (ETH) across all five seeds	rejected

Table 11: Machine-learning and reinforcement-learning results.

Too few independent samples. Overlapping h -day labels leave $n_{\text{eff}} \approx T/h$ independent observations: two years of daily data with $h = 7$ gives about 100. The Hanley–McNeil standard error of an AUC A near 0.5,

$$SE(\hat{A}) = \sqrt{\frac{A(1-A) + (n_1 - 1)(Q_1 - A^2) + (n_0 - 1)(Q_2 - A^2)}{n_1 n_0}}, \quad Q_1 = \frac{A}{2-A}, \quad Q_2 = \frac{2A^2}{1+A}, \quad (12)$$

is then about 0.058, so any AUC within 0.5 ± 0.11 is noise. Our daily models (0.40–0.51) all sit in that band; the hourly model (0.535, 95% interval 0.496–0.577) sits at its edge while the placebo reached 0.512–0.52. ETH’s random forest was significantly worse than chance (0.40, interval 0.34–0.47). When the signal-to-noise ratio is this low the Bayes error dominates and extra capacity only adds variance, which is why gradient boosting lost to the linear HAR model.

Reinforcement learning. Even with a generative model of the environment, learning an ε -optimal policy needs on the order of $\tilde{O}(|S||A|/((1-\gamma)^3\varepsilon^2))$ samples. With $|S||A| = 256$ and $\gamma = 0.95$ this is about $2 \times 10^6/\varepsilon^2$ transitions, against 365 a year of data. Q-learning’s convergence also needs stationary transition probabilities, which a regime-changing market violates. The agent memorised one bull year and held leverage into the 2022 crash.

The models learned the right economics (SHAP and partial-dependence plots show, for example, that the volatility model raises its forecast after downside semivariance), but with about three years of daily data they mostly fit noise. **No machine-learning or reinforcement-learning component is used by the final strategies.**

6 Strategy development and rejected approaches

6.1 Process discipline

Look-ahead can also enter through the researcher. These rules applied throughout; each experiment’s rules are recorded in the project’s evaluation protocol. 2024 under-performed as a validation year for the first candidates and then became training data. 2025 was used once as the test of an early candidate (K-01: –17.5% against BTC’s –7.6%); later candidates were compared on it before the final choice, so 2025 is not an untouched test for the final strategy, and we say so.

Fixed splits	train 2021–2023, validation 2024, test 2025
Rules before results	every experiment’s variants, gates and success criteria written down before it was run
Test once	a test period is evaluated once; a strategy is never changed after its test result
Validation becomes training	once a period has been used to select, it counts as training data
Ranges, not points	a parameter is accepted only if neighbouring values work too
Every trial logged	each backtest is appended to an experiment log; nothing is deleted
Luck benchmark	the deflated Sharpe ratio corrects for the number of trials (Section 10.4)

Table 12: Evaluation protocol.

6.2 BTC: from baselines to CTR-S

1. **Stages 1–9.** Baselines, trend ensembles, a regime gate and a chop filter led to K-01, frozen in protocol v6.
2. **Indicator research** (Section 4): risk and regime indicators are reliable, direction indicators are not.
3. **Stages 10–10b.** First regime-sizing designs. B7 reached Sharpe 1.13 and +375% but a –51% maximum drawdown.
4. **Stage 11.** The 2022 bear-market rallies were diagnosed as B7’s weakness and K-01’s soft 200-day gate was added. This design is CTR. It tied K-01 on Sharpe (1.1588 against 1.1556); CTR was chosen for +373% against +289% and 56% against 50% of quarters beating buy-and-hold, accepting a deeper drawdown.
5. **Stage 12.** The market-state framework. Refinements H1–H4 did not pass the cold-start gate, so CTR was unchanged.
6. **Stages 13–27.** A reliable-indicator baseline, a long-term direction study, a short-term layer, an ML risk engine, short-term and swing sleeves, and Bayesian regime-Kelly sizing. None beat CTR.
7. **Stages 28–31.** Rolling one-year windows over the full 2017–2026 history: CTR beat buy-and-hold in most bear windows but in only about 20% of bull windows.
8. **Stage 32.** Four pre-registered short sides on top of CTR (Section 7.4); only X3 passed every gate.
9. **Stage 33.** X3 was checked year by year and adopted as the BTC final, CTR-S. The short-borrow fee, first applied after the fact, moved into the engine; the 2021–24 result is unchanged at reported precision (+388.85% against +388.66%).

6.3 Rejected approaches and why they failed

Idea (stage)	Result	Why it failed
Fixed 3×ATR trailing stop (11)	fired 37 times; BTC Sharpe 1.16 → 0.71, return +373% → +125%	on daily crypto a price stop sells dips that recover, because jumps fade; under a random walk stops lower returns (Kaminski and Lo)

Idea (stage)	Result	Why it failed
No leverage after a 15% draw-down (11)	Sharpe 1.05, same maximum drawdown	lower Sharpe with no draw-down benefit
Smoother calm-trend label; downside-only brake (12)	Sharpe +0.01 / +0.02	within noise; failed the cold-start gate
Leverage only on full trend agreement (12)	Sharpe 1.13, deeper draw-down	worse on both counts
No leverage when far above the 200-day average, or on volume surges (12)	dropped before testing	the leveraged days in question earned more, not less
Short-term dip-buying in an uptrend, rule-based and ML (15)	−8.2% and −16.5% while BTC rose 102.5% (2022–24)	hit rate below break-even (Section 6.4)
ML/RL leverage control: quantile boosting, HMM, contextual bandit (16)	no better than the simple rules	no skill beyond a placebo
ML direction, triple-barrier filter, Q-learning	$AUC \leq 0.51$; Q-learning about −79%	inside the AUC noise band; RL sample complexity (Section 5)
Short-term and swing mean-reversion sleeves (25, 26)	edge per trade below the 0.30% round trip	cost hurdle
Bayesian regime-Kelly sizing (27)	no gain over fixed regime sizes	needs an estimate of μ_t , the least reliable input
1.5× above a banded 100-day average, else flat (31, 33)	maximum drawdown −79.5% over the longer history	−77% in the 2018 bear market alone
Volatility-targeted sizing (early stages)	average exposure 0.65–0.74	positions too small in bull markets; replaced by regime sizing
Intraday seasonality: long BTC 22:00–24:00 UTC	+2.1 bps a day, $t = 0.9$, negative in 2022	hour profiles anti-correlated across years: data snooping
Intraday time-series momentum	about 365 round trips a year	about 110% a year in costs
ETH/BTC pairs trading	ADF $p \geq 0.19$ in every year	one round trip of the ratio

6.4 Break-even hit rate for short-term trading

A trade with take-profit and stop at $\pm k\sigma$ and round-trip cost c has

$$\mathbb{E}[\text{PnL}] = pk\sigma - (1 - p)k\sigma - c = (2p - 1)k\sigma - c \implies p^* = \frac{1}{2} + \frac{c}{2k\sigma}. \quad (13)$$

With $k = 1$, daily $\sigma \approx 3\%$ and $c = 0.30\%$, $p^* = 0.55$. Our dip trades hit the target first in 24 of 49 decided cases ($\hat{p} = 0.49$), so they had to lose. The ML filter would have needed to lift p by 6 points, far beyond its AUC of 0.535.

ETH-specific idea	Result (2021–24 unless stated)	Stage
Slow direction alone (200-day, vote share, breakout, golden cross, momentum)	best Sharpe 0.79, below buy-and-hold's 0.87; drawdowns 53–62%	18
E3: 200-day + full $1.5\times$ when all votes agree + crash brake D1	Sharpe 0.95, +371%, but 6 of 16 quarters beating buy-and-hold, failed cold start; 2025 –11.2%	20, 20c
Fixes for E3 (hysteresis, cold-start rules, early entry)	all worse (Sharpe 0.59–0.80)	20b
Earlier ETH final G-06	2021–25 Sharpe 0.83, maximum drawdown –56.7%	4e, 20c
BTC short side on ETH	every variant lowered Sharpe and deepened the drawdown	32
The BTC framework unchanged (CTR-ETH)	2021–25 Sharpe 0.96, +482%, maximum drawdown –37.3%	12, 20c

Table 14: ETH candidates.

6.5 ETH: why no ETH-specific design survived

Crash speed against filter delay. A banded n -day state exits only when $P_t < \text{SMA}_n(P)_t(1 - b_t)$, and the SMA lags by $(n - 1)/2$ days. From its 8 November 2021 top ETH fell 33.5% before the 200-day state exited on 7 January 2022, so slow-only designs carried most of each crash (drawdowns 53–62%).

The crash indicator D1 and regime concentration. D1 is the ratio of a 72-hour EWMA of hourly downside semivariance to its trailing 90-day median,

$$D1_t = \frac{\text{EWMA}_{72h}(r_h^2 \mathbf{1}\{r_h < 0\})_t}{\text{median}_{90d}(\text{EWMA}_{72h}(r_h^2 \mathbf{1}\{r_h < 0\}))_t}. \quad (14)$$

In its top decile next week's mean maximum drawdown was –11.0% against –7.4%, and it caught 22% of the worst-decile weeks (a lift of 2.2), but beyond current volatility its rank correlation with next-week downside volatility is about zero (–0.04, +0.06, –0.19, –0.16 by year): it is a thermometer, not a forecast, and the storm brake already uses the same mechanism. E3 reached Sharpe 0.95, but removing its ability to lever up in the first 200 days cut its return from +371% to +171%:

$$R_{E3} = R_{\text{early 2021}} + R_{\text{rest}}, \quad R_{\text{early 2021}}/R_{E3} \text{ large}, \quad (15)$$

a sign of performance concentrated in one regime. E3 then lost 11.2% in 2025 while CTR-ETH made 54.9%.

7 The final strategy

7.1 The market-state framework

`src/strategies/market_state.py` reads any dataset the same way. From data up to each daily close it computes the trend votes, the 200-day gate, the volatility percentile, trend quality, the downside risk forecast and the squeeze flag, and names the market's state. Every threshold is relative to the data's own trailing history, and averages warm up adaptively, so the same code works on a new coin, period or price level from a cold start.

State	Definition (priority order)	BTC			ETH		
		days	spell	long	days	spell	long
Storm	risk forecast $> 1.5 \times$ its 1-year median	4%	7 d	0.15	7%	11 d	0.10
Downtrend	gate down, $<$ half the votes up	34%	31 d	0.04	34%	27 d	0.04
Bear rally	gate down, $\geq$ half the votes up	5%	9 d	0.38	7%	11 d	0.40
Calm uptrend	gate up, votes up, calm and orderly	20%	7 d	1.42	15%	10 d	1.38
Volatile up-trend	gate up, votes up, stormy	9%	6 d	0.54	12%	9 d	0.52
Uptrend	gate up, votes up, neither	17%	7 d	0.79	17%	10 d	0.84
Fading uptrend	gate up, $<$ half the votes up	10%	10 d	0.27	9%	11 d	0.26

Table 15: Market states, 2021–2025: share of days, average spell length and average long exposure. States are persistent: unchanged the next day 83–97% of the time.

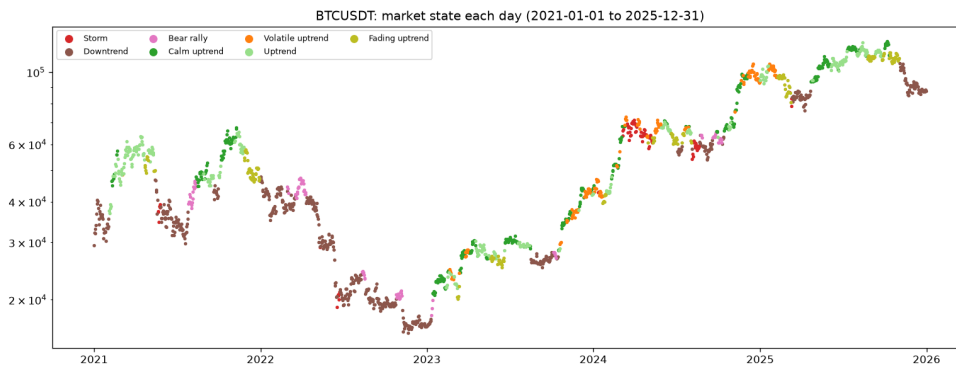


Figure 17: BTC 2021–2025, each day coloured by its market state.

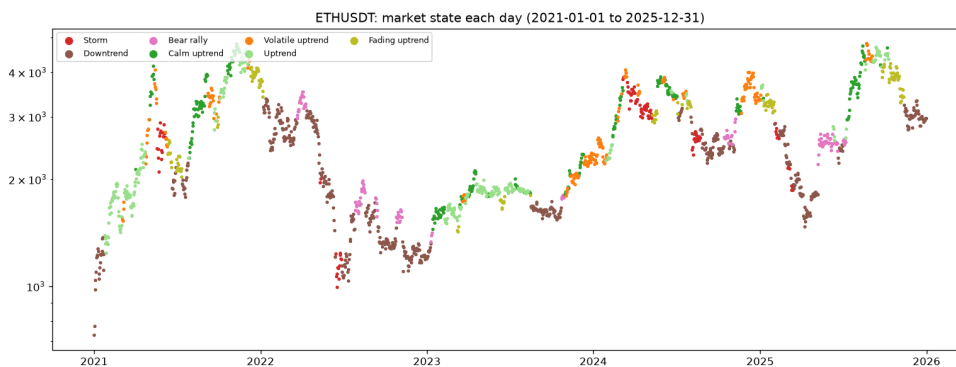


Figure 18: ETH 2021–2025, each day coloured by its market state.

7.2 The daily decision

Evaluated at each daily close; the order fills at the next day's open.

1. Direction: seven banded trend votes. For each $n \in \{20, 30, 50, 75, 100, 150, 200\}$, with daily volatility $\sigma_t = \sqrt{\text{EWMA}_{60}(r_t^2)}$ as a hysteresis band,

$$V_{n,t} = \begin{cases} 1 & P_t > \text{SMA}_{n,t}(1 + \sigma_t) \\ 0 & P_t < \text{SMA}_{n,t}(1 - \sigma_t) \\ V_{n,t-1} & \text{otherwise,} \end{cases} \quad \text{votes}_t = \frac{1}{7} \sum_n V_{n,t}. \quad (16)$$

2. Regime. With π_t the rank of 14-day Parkinson volatility within the trailing 365 days, ER_t the 30-day efficiency ratio and CI_t the 14-day chopiness index,

$$\pi_t = \frac{1}{365} \sum_{s=t-365}^{t-1} \mathbf{1}\{\sigma_s^P < \sigma_t^P\}, \quad (17)$$

$$\text{regime}_t = \begin{cases} 1.5 & (\pi_t \leq 0.5 \wedge ER_t > \text{median}_{s \leq t} ER_s) \vee CI_t < 38.2 \\ 0.6 & \pi_t > 0.5 \\ 1 & \text{otherwise.} \end{cases} \quad (18)$$

3. Storm brake. From hourly data,

$$F_t = \sqrt{365 \cdot \text{EWMA}_{30}(0.5 BV_t + RS_t^-)}, \quad \text{brake}_t = \begin{cases} 0.5 & F_t > 1.5 \cdot \text{median}_{365}(F)_t \\ 1 & \text{otherwise.} \end{cases} \quad (19)$$

4. Squeeze. $\text{squeeze}_t = 0.75$ while the Bollinger Bands $\text{SMA}_{20} \pm 2\text{sd}_{20}$ sit inside the Keltner Channel $\text{EMA}_{20} \pm 1.5 \text{ATR}_{20}$, else 1.

5. Long-term gate. $\text{gate}_t = 0.5 + 0.5 V_{200,t}$.

Target.

$$\boxed{\text{Long}_t = \min(\text{votes}_t \cdot \text{regime}_t \cdot \text{brake}_t \cdot \text{squeeze}_t, 1.5) \cdot \text{gate}_t} \quad (20)$$

as a fraction of equity. Every parameter (the 20–200-day averages, the 0.5 median split, Chopiness 38.2, $1.5 \times$ leverage, the 30-day EWMA, half-size shorts, the -60% capitulation line) is a round or conventional value fixed before testing; none was optimised.

7.3 Why each layer is there

The vote ensemble. If K signals have common variance σ^2 and pairwise correlation ρ , their average has variance $\sigma^2(\rho + \frac{1-\rho}{K})$: 74% of a single signal's for $K = 7$ and $\rho \approx 0.7$. More important, the vote share moves in steps of $1/7$, so turnover and cost are a fraction of a binary rule's, and the position mixes group delays from 9.5 to 99.5 days instead of betting on one lag. This is why the result is insensitive to the exact lookbacks while a single 30-day rule is not.

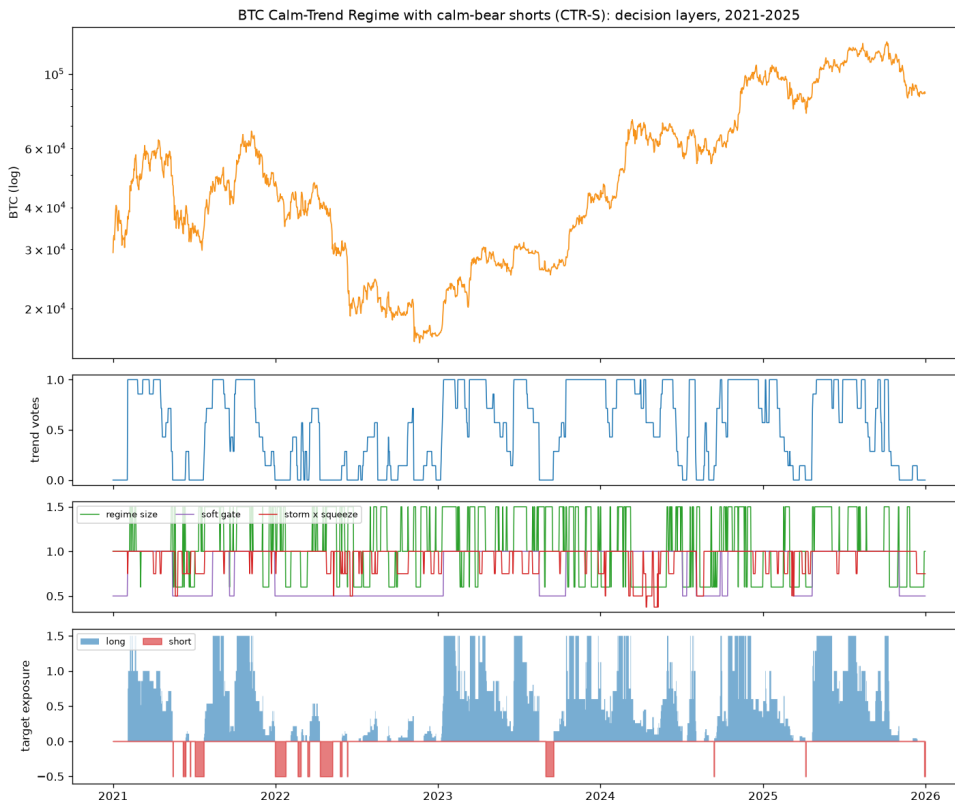
Proposition 2 (Rank thresholds calibrate the rule to each coin). Let X_t be any risk measure and $\pi_t(X) = \frac{1}{N} \sum_{s=t-N}^{t-1} \mathbf{1}\{X_s < X_t\}$. For any strictly increasing g , $\pi_t(g(X)) = \pi_t(X)$, because $g(X_s) < g(X_t) \Leftrightarrow X_s < X_t$ term by term.

Layer	Evidence (2021–24)	Without it
7 averaged votes	single 30-day rule: Sharpe 1.08; its 20- and 45-day neighbours 0.77 and 0.68	whipsaw: 50 trades under 30 days lost 7,900 USDT
Calm-trend 1.5×	trend payoff positive in calm trends in 4 of 4 years; the 18.5% of days above 1× earned most of the return	long-term votes alone: Sharpe 0.96, +252%
Stormy 0.6×	momentum payoff about zero or negative in high volatility	stormy entries lost 3,700 USDT net
Downside storm brake	sell-off volatility persists about twice as much as rally volatility	a brake on all volatility also fires in rallies
Squeeze 0.75×	volatility rises 10–14% after a squeeze	Sharpe 1.13 → 1.04
200-day soft gate	2022’s bear rallies were the ungated design’s main loss (−51% drawdown)	2022: −34% instead of −16%

Table 16: Ablation evidence for each layer.

“Stormy” therefore means stormy for that coin at its own volatility level (85% for ETH, 65% for BTC) with no free parameter, and the same holds for the efficiency ratio against its own median and the storm brake against its own median. This is why the same code is the ETH strategy, and why nothing was tuned for ETH.

Kelly and the leverage cap. The growth-optimal exposure $f_t^* = \mu_t / \sigma_t^2$ is largest in calm trends, where $\mu_t > 0$ in 4 of 4 years and σ_t is below its median. The 1.5× cap is a fractional-Kelly guard against estimation error in μ_t , the least reliable quantity in the system.

**Figure 19:** BTC 2021–2025: price, vote share, sizing layers and the resulting target exposure.

7.4 The BTC short side (CTR-S)

CTR-S holds $-0.5\times$ equity only while all five conditions hold, and covers at the next open as soon as one fails: Four variants were written down before any was run, with gates set in advance:

	Condition	Why
a	the long target is 0	the short never overrides or reduces a long
b	the 200-day gate is down	a bear market, not a dip in a bull market
c	all 7 trend votes are down	every horizon agrees; the first vote to turn up (usually the 20-day) covers
d	volatility percentile ≤ 0.5	orderly declines persist; violent markets squeeze shorts
e	BTC less than 60% below its 365-day high	after capitulation rebounds are sharpest: negative extremes mean-revert more strongly (Corbet and Katsiampa)

Table 17: Short-side conditions: the calm, fully confirmed, non-capitulation part of the Downtrend state, about 6% of days.

beat CTR's Sharpe and return on 2021–24 also at double costs, keep 80% of the Sharpe at every neighbouring setting, win at least as many rolling one-year windows as CTR, and have more positive bear-market windows. Over 2021–2025 the short side turned 2022 from -16.1% into

Variant	Short side	Sharpe	Return	Max DD	Result
Long-only CTR	none	1.159	+373%	−37.5%	baseline
X1	$-0.5\times$ when a–d hold	1.10	+343%	−33.3%	failed the Sharpe gate
X2	$-1.0\times$ when a–d hold	0.98	+284%	−36.7%	failed
X3 = CTR-S	$-0.5\times$ when a–e hold	1.163	+389%	−29.6%	passed every gate
X4	$-0.5\times$ without condition d	0.95	+261%	−40.3%	failed

Table 18: Pre-registered short-side variants, 2021–2024.

-1.1% , cut the maximum drawdown from -37.5% to -29.6% and the longest time under water from 723 to 496 days. It costs a few points in bull years (2021 -13 , 2023 -4 , 2024 -5 points) from shorts caught by fast reversals. The 15 short trades sum to $+3.8\%$ of equity but net -334 USDT, because the 2022 gains came at low equity; their value is the path, since the 2023–24 bull market compounds from a higher base.

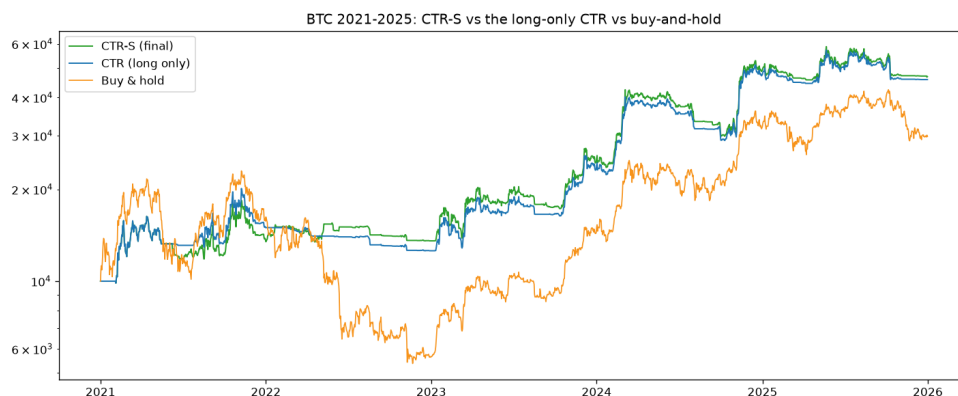


Figure 20: BTC 2021–2025: CTR-S with shorts, the long-only CTR and buy-and-hold (log scale).

7.5 ETH (CTR-ETH)

CTR-ETH applies the same rules to ETH, long only. ETH's direction is as unpredictable as BTC's, its volatility is more predictable (HAR $R^2 = 0.73$), and trend-following paid in calm ETH trends in 4 of 4 years. The fast votes leave within days of a break, so exposure falls stepwise (6/7, 5/7, ...) long before the slow state exits; this is why CTR-ETH's maximum drawdown is -37% while slow-only designs had about -58% .

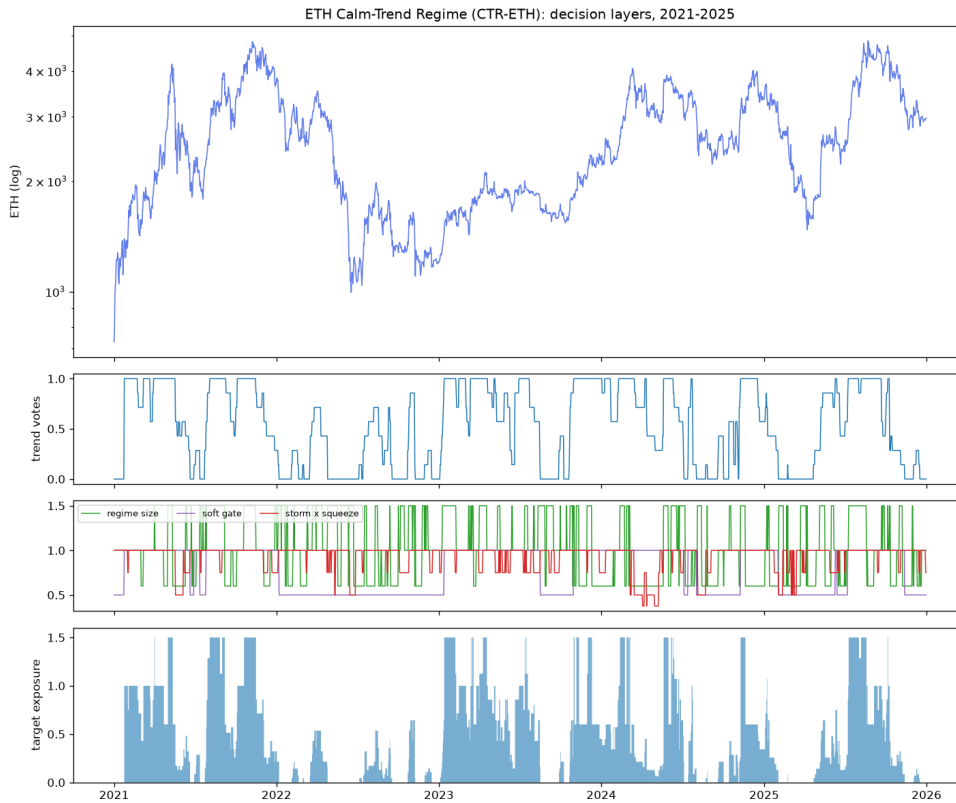


Figure 21: ETH 2021–2025: price, vote share, sizing layers and target exposure.

7.6 Execution settings

Fee + slippage	0.15% of notional per fill	a round trip costs 0.30%
Fill price	next bar's open	the engine applies the lag
Maximum long / short	1.5× / 0.5× equity	short for BTC only
Borrowed USDT	8% a year, daily	when the long exceeds equity
Borrowed BTC	10% a year, daily, on the short's value	assumed margin rate
Rebalance band	resize only if the target moves > 0.25	limits churn and cost
Re-entry cool-down	5 days	after an exit, before re-entering the same side

Table 19: Execution settings ([src/strategies/execution.py](#)).

8 Backtest engine and look-ahead control

8.1 Design

The engine’s one job is to answer honestly what a strategy would have earned trading only on information it had, after paying 0.15% on every fill; every design choice favours correctness and pessimism over speed. Strategies compute their signals with vectorised pandas code and the engine executes them in a plain bar loop ([src/backtest/engine.py](#)). A strategy returns a target position per bar; the engine, not the strategy, applies the one-bar lag. A strategy is a position process w_t that must be $\mathcal{F}_t$ -adapted, and the portfolio return is

$$R_{t+1} = w_t r_{t+1}^{\text{open}} - c |w_t - w_{t-1}| - b(w_t - 1)^+, \quad c = 0.0015, \quad (21)$$

with b the daily borrowing rate on leverage above 1.

Event	Fill rule	Why
Entry, exit, reversal	next tradable bar’s open	the earliest price available after the decision
Stop or trailing stop	at the stop if touched; at the open if the bar opens beyond it	the data has real gaps and 10–20% wicks
Stop and target in one bar	stop assumed first	OHLC cannot tell which came first
Trailing level	best price up to the previous bar	avoids assuming the high came before the low
Outage bar	no fills; orders wait	<code>is_filled</code> or <code>trades == 0</code>
End of data	closed at the last close, with a fee	every trade is closed and costed

Table 20: Fill rules. Sharpe and Sortino use daily equity returns $\times \sqrt{365}$ with a zero risk-free rate; buy-and-hold enters at the first tradable open and pays the same costs.

8.2 Keeping look-ahead out

- **Features.** Only trailing windows, `ewm`, non-negative shifts, expanding statistics and percentiles within trailing windows. Forbidden in any strategy: centred windows, back-fill, interpolation, two-sided filters, Kalman smoothers, smoothed HMM probabilities, zigzag labelling, Ichimoku’s backward-shifted lines, and any full-sample statistic applied to earlier points.
- **Data.** Gaps are filled forward only; realised measures for a day use only that day’s hours; an unfinished last day is dropped before any decision.
- **Signal truncation test** ([src/backtest/checks.py](#)). Signals are recomputed on data cut at six points between bar 300 and the end and must be identical to the full-data signals up to each cut; otherwise the test fails with the first differing timestamp.
- **Backtest truncation test.** The backtest re-run up to a cut must reproduce every trade and equity value before the cut.
- **Engine validation.** A hand-computed round trip (967.05 USDT), gap and same-bar tests, outage tests, the invariants $\text{equity} = \text{cash} + \text{position value}$ and $\sum \text{trade PnL} = \text{equity change}$ on real data, an always-long strategy reproducing buy-and-hold exactly, random signals on zero-drift prices costing exactly 0.30% per round trip, and a deliberately cheating `shift(-1)` signal that the causality check catches.

Both truncation tests run in the test suite for both strategies and every indicator, and inside every run on new data.

9 Risk management

Control	BTC (CTR-S)	ETH (CTR-ETH)
Position size	$\leq 1.5\times$ long, only in calm trends; $\leq 0.6\times$ when stormy; $\leq 0.5\times$ short	$\leq 1.5\times$, only in calm ETH trends; never short
Sell-off protection	storm brake on downside volatility (twice the forecasting weight of upside)	storm brake on ETH's own hourly sell-off volatility, squeeze, 200-day gate
Long exit	trend-break stop: the position shrinks vote by vote and closes when all votes are down	same
Short exit	cover at the next open when any of a–e fails, usually the 20-day vote turning up	–
Long and short	never at once: the short exists only when the long target is exactly zero (tested)	–
Joint risk	ETH crashes with BTC on 82% of the worst days; run together, the two strategies are one risk position in a crash.	

Table 21: Risk controls.

9.1 Payoff, not accuracy

With win rate p , average win W and average loss L as fractions of equity,

$$\mathbb{E}[\text{trade}] = pW - (1 - p)L, \quad p^* = \frac{L}{W + L}. \quad (22)$$

2021–2025	BTC (36 trades)	ETH (21 trades)
Win rate	33.3%	23.8%
Average win / average loss (equity)	+22.0% / –2.2%	+56.0% / –2.4%
Payoff ratio	10.0 : 1	23.4 : 1
Profit factor	4.31	5.06
Break-even win rate p^*	9.1%	4.1%
Largest losing trade (equity)	–10.6% (5 August 2024 crash)	–8.3%
Longest winning trade	290 days	258 days

Table 22: Risk–reward measured on equity. Many small losses are cut quickly and a few large wins run for months.

9.2 Drawdown arithmetic and volatility drag

Recovering from a loss L needs a gain of $L/(1 - L)$, and geometric growth satisfies $g \approx \mu - \frac{1}{2}\sigma^2$. At BTC's 65% volatility the drag is about 21% a year at full exposure and at ETH's 85% about 36%; halving exposure in storms cuts the drag on those days by three quarters. This asymmetry, not better prediction, is why both strategies end well ahead while capturing less of each bull market.

9.3 Tail risk and known risks

- **Shocks from calm markets** hit while the strategies hold up to $1.5\times$.

Series	Maximum drawdown	Gain to recover
BTC buy-and-hold	−76.6%	+327%
BTC CTR-S	−29.6%	+42%
BTC long-only CTR	−37.5%	+60%
ETH buy-and-hold	−79.3%	+383%
ETH CTR-ETH	−37.3%	+60%

Table 23: The asymmetry of losses.

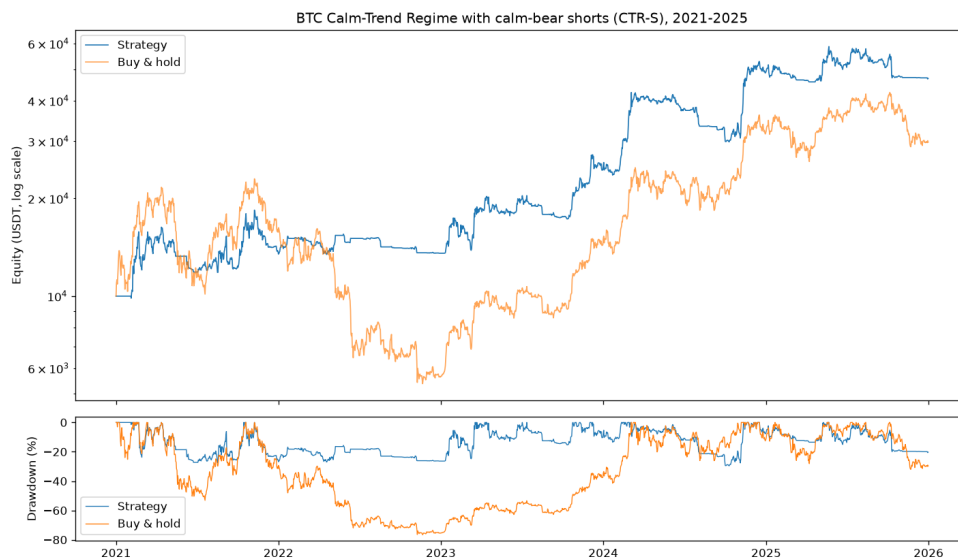
	BTC	ETH
Worst day	−15.7% (7 Sep 2021)	−18.6% (7 Sep 2021)
Worst 7 days	−23.8%	−25.7%
Daily 95% VaR / CVaR	−2.6% / −4.6%	−3.5% / −5.9%
Deepest drawdown and recovery	−29.6%; longest under water 496 days	−37.3%; 828 days to recover

Table 24: Measured tail risk, 2021–2025.

- **Short squeezes in bull-market dips:** June 2021 (−5.9% of equity), September 2023 (−2.0%), September 2024 (−2.2%).
- **Lag in bull markets:** 2021 Q1 BTC +52% against +103%; ETH 2021 +175% against +399%.
- **Cold start on ETH is weaker:** from scratch in 11 quarterly windows its mean Sharpe was 0.62 against buy-and-hold's 0.74.
- **Borrow cost** for BTC is an assumed 10% a year; shorts were held on about 6% of days and cost 235 USDT over 2021–25.

10 Results, 2021–2025

One continuous run per coin: 10,000 USDT, 0.15% per fill, next-open execution, financing included.

**Figure 22:** BTC: CTR-S equity and drawdown against buy-and-hold (log scale).

Required metric	BTC (CTR-S)	ETH (CTR-ETH)
Gross profit (USDT)	47,963.39	60,041.03
Net profit (USDT)	36,839.51	48,168.68
Total closed trades	36 (21 long, 15 short)	21
Win rate	33.33%	23.81%
Maximum drawdown	−29.56%	−37.34%
Gross loss (USDT)	−11,123.88	−11,872.35
Average winning trade (USDT)	3,996.95	12,008.21
Average losing trade (USDT)	−463.49	−742.02
Buy-and-hold return	+197.92%	+306.46%
Largest losing trade (USDT)	−3,950.76	−3,360.95
Largest winning trade (USDT)	19,827.38	24,676.80
Sharpe ratio	0.98	0.96
Sortino ratio	1.60	1.53
Average holding duration	41.6 days	64.3 days
Maximum holding duration	290 days	258 days
<i>Total / annualised return</i>	<i>+368.40% / 36.16%</i>	<i>+481.69% / 42.19%</i>
<i>Calmar ratio</i>	<i>1.22</i>	<i>1.13</i>
<i>Exposure; fees; financing (USDT)</i>	<i>82%; 4,412; 1,194</i>	<i>74%; 3,223; 753</i>
<i>Sharpe at double costs</i>	<i>0.90</i>	<i>0.91</i>
<i>Buy-and-hold Sharpe / maximum drawdown</i>	<i>0.67 / −76.63%</i>	<i>0.75 / −79.30%</i>

Table 25: The 15 required metrics and extras ([results/btc/2021-2025/metrics.md](#), [results/eth/2021-2025/metrics.md](#)).

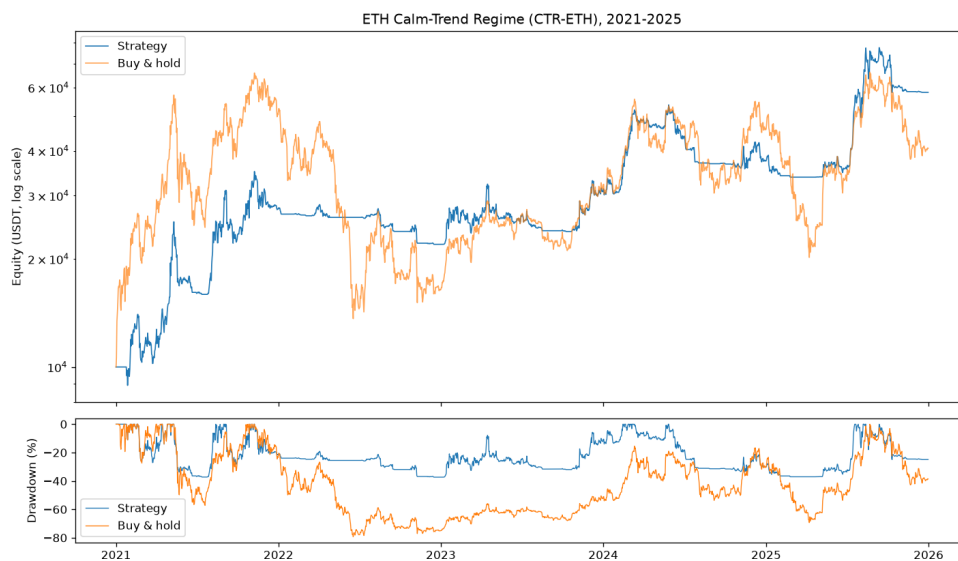


Figure 23: ETH: CTR-ETH equity and drawdown against buy-and-hold (log scale).

10.1 Year by year

Year	BTC				ETH		
	CTR-S	CTR	BTC	CTR-S DD	CTR-ETH	ETH	DD
2021	+37.1%	+49.9%	+59.8%	−27.4%	+175.0%	+399.2%	−37.2%
2022	−1.1%	−16.1%	−64.2%	−12.7%	−20.2%	−67.5%	−22.5%
2023	+80.8%	+84.4%	+155.6%	−20.5%	+40.0%	+90.8%	−26.2%
2024	+99.6%	+104.1%	+121.3%	−29.6%	+21.8%	+46.3%	−33.6%
2025	−4.3%	−3.3%	−6.3%	−20.6%	+55.4%	−11.0%	−25.0%

Table 26: One continuous run per coin, cut by calendar year ([results/btc/2021-2025/yearly.csv](#)).

10.2 Quarter by quarter

Table 27: Quarterly returns ([results/btc/2021-2025/quarterly.csv](#), [results/eth/2021-2025/quarterly.csv](#)). Bold: the strategy beat buy-and-hold. CTR-S beat BTC in 11 of 20 quarters and in 8 of the 9 where BTC fell; CTR-ETH beat ETH in 10 of 20 and in 7 of the 8 where ETH fell.

Quarter	CTR-S	BTC	CTR-ETH	ETH
2021Q1	+52.1%	+103.1%	+16.2%	+160.6%
2021Q2	− 22.0%	−40.3%	+38.9%	+18.6%
2021Q3	+3.0%	+25.1%	+40.5%	+31.9%
2021Q4	+12.2%	+5.5%	+21.3%	+22.5%
2022Q1	+5.8%	−1.5%	− 0.3%	−10.7%
2022Q2	+3.8%	−56.2%	− 4.8%	−67.4%
2022Q3	−6.5%	−2.6%	−8.6%	+24.1%
2022Q4	− 3.7%	−14.8%	− 8.0%	−10.0%
2023Q1	+38.0%	+72.1%	+20.7%	+52.3%
2023Q2	+5.5%	+7.0%	−0.6%	+6.2%
2023Q3	− 11.3%	−11.5%	− 9.0%	−13.6%
2023Q4	+39.9%	+56.8%	+28.2%	+36.6%
2024Q1	+68.7%	+68.6%	+59.6%	+59.8%
2024Q2	− 8.5%	−11.9%	−10.5%	−5.7%
2024Q3	−17.4%	+0.9%	− 16.0%	−24.3%
2024Q4	+56.6%	+47.8%	+1.5%	+28.3%
2025Q1	− 5.7%	−11.8%	− 9.8%	−45.4%
2025Q2	+11.3%	+29.8%	+6.4%	+36.4%
2025Q3	−0.3%	+6.4%	+89.7%	+66.8%
2025Q4	− 8.5%	−23.1%	− 14.7%	−28.3%

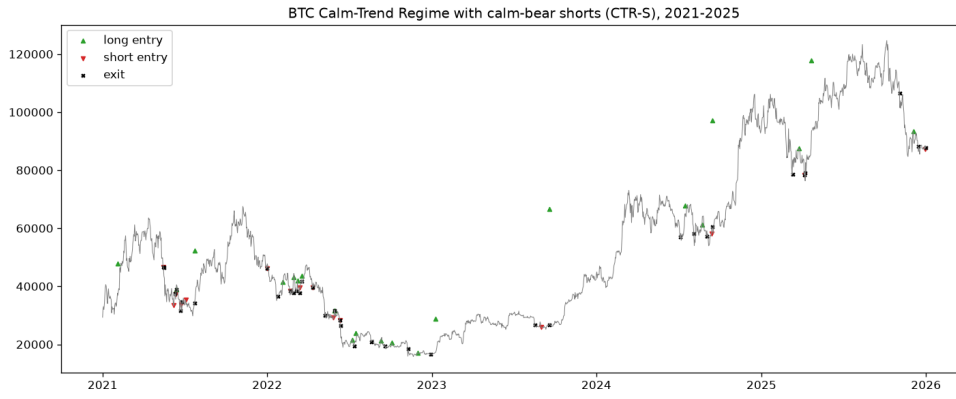


Figure 24: BTC trades on price, 2021–2025.

Check	BTC CTR-S	ETH CTR-ETH
Neighbouring settings (2021–24)	long side keeps $\geq 96\%$ of the Sharpe (high-vol size, volatility threshold, storm brake, gate floor and gate lookback each moved both ways); worst short-side neighbour 1.14 against 1.16	$\geq 90\%$; worst 0.81 against 0.90 (gate floor 0.7)
Double costs (0.30% per fill)	Sharpe 1.08 (2021–24), 0.90 (2021–25)	0.85, 0.91
Daily candles only	1.17 against 1.16	0.88 against 0.90
Cold start, 11 quarterly windows	mean Sharpe 1.49; drawdowns -28% to -38% ; beat buy-and-hold's Sharpe in 7	mean 0.62 against buy-and-hold's 0.74; beat it in 5

Table 28: Robustness checks.

10.3 Robustness

10.4 How much of the result could be luck?

For N independent zero-skill strategies tested over T years the expected best Sharpe ratio is (Bailey and López de Prado)

$$\mathbb{E}[\max_{i \leq N} \widehat{SR}_i] \approx \frac{1}{\sqrt{T}} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right], \quad \gamma \approx 0.5772, \quad (23)$$

which at $T = 4$ is 0.79, 1.27 and 1.63 for $N = 10, 100$ and 1,000. About 1,000 BTC backtests were logged, but they are small variations of about a dozen design families, so the effective N is 10–20 and the luck benchmark about 0.8–1.0. The case for the strategies therefore does not rest on the 2021–24 Sharpe ratio alone but on stable neighbours, consistency year by year and the cold-start runs, none of which a lucky pick produces.

11 Limitations and further work

- **Bull markets.** The strategies hold about $0.5\text{--}0.8\times$ on average in bull years and lag strong rallies; they beat buy-and-hold over full cycles, not every year.
- **Calm-market crashes.** Jumps often arrive from calm markets, where exposure can be $1.5\times$.
- **Sample size of the short side.** It traded 15 times in 2021–25; its gain rests mainly on one bear market.

- **Correlation.** ETH and BTC crash together, so the two strategies are not diversified in a crash.
- **2025 was seen.** It was used once as an early candidate's test and then for comparing candidates, so it is not an untouched test.

Data that would improve the analysis: perpetual futures funding rates and open interest, liquidation data, minute or tick trades and order-book depth, other exchanges, options implied volatility (Deribit DVOL), macro calendars, on-chain stablecoin and exchange flows, and pre-2021 history.

References

1. Anis, A., Lloyd, E. (1976). The expected value of the adjusted rescaled Hurst range of independent normal summands. *Biometrika*.
2. Azar, M., Munos, R., Kappen, H. (2013). Minimax PAC bounds on the sample complexity of reinforcement learning with a generative model. *Machine Learning*.
3. Bailey, D., López de Prado, M. (2014). The deflated Sharpe ratio. *Journal of Portfolio Management*.
4. Barndorff-Nielsen, O., Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*.
5. Barndorff-Nielsen, O., Kinnebrock, S., Shephard, N. (2010). Measuring downside risk: realised semi-variance. In *Volatility and Time Series Econometrics*.
6. Corbet, S., Katsiampa, P. (2020). Asymmetric mean reversion of Bitcoin price returns. *International Review of Financial Analysis*.
7. Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*. doi:10.1093/jjfinec/nbp001
8. Diebold, F., Mariano, R. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*.
9. Engle, R., Granger, C. (1987). Co-integration and error correction. *Econometrica*. doi:10.2307/1913236
10. Glosten, L., Jagannathan, R., Runkle, D. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*.
11. Grinold, R. (1989). The fundamental law of active management. *Journal of Portfolio Management*.
12. Hamilton, J. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*.
13. Hanley, J., McNeil, B. (1982). The meaning and use of the area under a ROC curve. *Radiology*.
14. Hill, B. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*. doi:10.1214/aos/1176343247
15. Joe, H. (1997). *Multivariate Models and Dependence Concepts*. Chapman & Hall.
16. Kaminski, K., Lo, A. (2014). When do stop-loss rules stop losses? *Journal of Financial Markets*.
17. Kaufman, P. (1995). *Smarter Trading*. McGraw-Hill.
18. Kelly, J. (1956). A new interpretation of information rate. *Bell System Technical Journal*.
19. Lo, A., MacKinlay, C. (1988). Stock market prices do not follow random walks: evidence from a simple specification test. *Review of Financial Studies*. doi:10.1093/rfs/1.1.41
20. López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
21. Moreira, A., Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*. doi:10.1111/jofi.12513
22. Moskowitz, T., Ooi, Y. H., Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*. doi:10.1016/j.jfineco.2011.11.003
23. Page, E. (1954). Continuous inspection schemes. *Biometrika*.
24. Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*.
25. Patton, A., Sheppard, K. (2015). Good volatility, bad volatility: signed jumps and the persistence of volatility. *Review of Economics and Statistics*.
26. Watkins, C., Dayan, P. (1992). Q-learning. *Machine Learning*.
27. Zarattini, C., Pagani, A., Barbon, A. (2025). Catching crypto trends. SSRN working paper.

The full annotated list, with what we took from each source, is in [reports/glossary.md](#).